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Accelerating hybrid ANN–ODE frameworks using surrogate machine learning and metaheuristic optimization for predicting recurrent venous thromboembolism

Mohamad Al Bannoud · Tiago Dias Martins · Silmara Aparecida de Lima Montalvão · Joyce Maria Annichino-Bizzacchi · Rubens Maciel Filho · Maria Regina Wolf Maciel

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Abstract

Accurate and computationally efficient modeling of complex biological processes remains a major challenge in personalized medicine. This study introduces a hybrid framework that replaces the computationally intensive ordinary differential equation (ODE) system in hybrid artificial neural network–ODE (ANN–ODE) models with machine learning (ML) surrogate models. Using clinical and hematological data, the framework estimates patient-specific kinetic parameters of the coagulation cascade and predicts the endogenous thrombin potential (ETP), which serves as a discriminative threshold for classifying recurrent venous thromboembolism (RVTE) risk. Fourteen ML algorithms and eight metaheuristic optimization algorithms (MOAs) were systematically evaluated to identify optimal model–optimizer combinations. Among all candidates, the artificial neural network (ANN) acts as the best surrogate, yielding a root mean squared error (RMSE) below 0.3 for both the training and test sets. The ANN coupled with the Grey Wolf Optimizer (ANN–GWO) achieved the best performance, maintaining a relative accuracy of 97.97% compared with full ODE simulations, while reducing optimization time by over 99%. Particle swarm optimization (PSO) also exhibited competitive performance, confirming the robustness of swarm-based search strategies. These results demonstrate that replacing mechanistic ODE systems with machine-learning surrogates can substantially reduce computational complexity without sacrificing predictive accuracy. The proposed ANN–GWO and ANN–PSO models provide efficient, accurate, and scalable computational tools for RVTE prediction and personalized medicine. The findings support the use of ANNs as surrogate models and highlight GWO and PSO as robust and reliable optimizers for RVTE prediction within the tested hybrid framework. This guidance is particularly relevant for further extensions and improvements of the hybrid framework that require low computational cost without compromising accuracy, such as backward and forward selection of clinical features and kinetic parameters.

Keywords Recurrent venous thromboembolism · Surrogate machine learning · Blood coagulation cascade · Metaheuristic optimization

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Abbreviations

ABC	Artificial Bee Colony
ACC	Accuracy
ACO	Ant Colony Optimization
Adam	Adaptive Moment Estimation
ANN	Artificial Neural Network
ANOVA	Analysis of Variance
AT-III	Antithrombin III
AUC	Area Under the Curve
BA	Bat Algorithm
BDF	Backward Differentiation Formula
BR	Bayesian Ridge Regression
CatBoost	Categorical Boosting
CI	Confidence Interval
CTESN	Continuous-Time Echo State Networks
DCA	Decision Curve Analysis
DMP	Discriminant Margin Potential
DT	Decision Tree
DVT	Deep Vein Thrombosis
ETP	Endogenous Thrombin Potential
EN	ElasticNet Regression
FA	Firefly Algorithm
GA	Genetic Algorithm
GB	Gradient Boosting
GWO	Grey Wolf Optimizer
KNN	K-Nearest Neighbors
LightGBM	Light Gradient Boosting Machine
Lasso	Lasso Regression
LR	Linear Regression
MAD	Mean Absolute Deviation
MAE	Mean Absolute Error
ML	Machine Learning
MLP	Multilayer Perceptron
MOA	Metaheuristic Optimization Algorithm
ODE	Ordinary Differential Equation
PE	Pulmonary Embolism
PSO	Particle Swarm Optimization
QSP	Quantitative Systems Pharmacology
RBC	Red Blood Cell
RF	Random Forest
RMSE	Root Mean Squared Error
RMSProp	Root Mean Square Propagation
ROC	Receiver Operating Characteristic
RR	Ridge Regression
RVTE	Recurrent Venous Thromboembolism
SGD	Stochastic Gradient Descent
SM	Supplementary Material

SVR	Support Vector Regression
TNR	True Negative Rate
TPR	True Positive Rate
VTE	Venous Thromboembolism
WBC	White Blood Cell
WOA	Whale Optimization Algorithm
XGBoost	eXtreme Gradient Boosting

1 Introduction

Thrombosis is characterized by the formation of abnormal blood clots within blood vessels, potentially obstructing normal blood flow. This pathological process underlies several severe conditions, including deep vein thrombosis (DVT), pulmonary embolism (PE), myocardial infarction, and ischemic stroke [1]. Thrombosis and its complications represent a primary global health concern, contributing to significant morbidity, mortality, and socioeconomic burden [2, 3]. Following an initial thrombotic event, a primary clinical concern is recurrence, influenced by factors such as inadequate anticoagulant therapy, malignancy, prothrombotic disorders, and poor treatment adherence [4, 5]. Accurate prediction of recurrent thrombosis is thus essential for guiding clinical decisions, individualizing therapy, and improving long-term outcomes.

Various clinical risk scores have been developed and are routinely used in medical practice to predict venous thromboembolism (VTE), including the Khorana [6], Caprini [7], and Wells [8, 9] scores. For recurrent venous thromboembolism (RVTE), scores such as the Ottawa [10], DASH [11], and Vienna [12] have been proposed. Despite widespread use, these tools present several limitations: reliance on a restricted set of variables, limited adaptability to complex clinical scenarios, low specificity, and reduced applicability across diverse populations [13–15]. As an alternative, data-driven models, particularly those based on machine learning (ML), offer several advantages. These include the ability to incorporate a broader range of clinical, laboratory, and demographic variables, improved predictive accuracy, and the capacity to identify complex, nonlinear, and high-dimensional patterns that often elude traditional statistical approaches [16–18]. Such models show strong potential for enhanced risk stratification and personalization of care in RVTE [19] and other complex conditions [20, 21].

A key barrier to clinical adoption of ML-based models for RVTE prediction is their black-box nature, which limits interpretability and hinders trust among healthcare professionals [22]. To address this, our group has proposed a hybrid framework that integrates data-driven techniques with the mechanistic component of the blood coagulation process [23]. In this framework, clinical and hematological patient data are used to estimate personalized kinetic parameters of the coagulation cascade, which vary substantially among models reported in the literature [24]. These parameters are then used to solve a system of ordinary differential equations (ODEs) modeling the coagulation cascade. From the resulting thrombin generation curve, the endogenous thrombin potential (ETP) is calculated and used, via a predefined optimal threshold, to classify patients as high or low risk for RVTE. This hybrid model poses its challenges. It combines a multilayer perceptron (MLP) to map clinical variables to kinetic parameters with a mechanistic ODE-based computation of ETP. Because the whole system is non-differentiable, traditional gradient-based optimization is not feasible, necessitating the use of gradient-free optimizers, such as metaheuristic algorithms, which do not require differentiability but typically require many function evaluations [25].

Despite their advantages, hybrid mechanistic–ML models face computational challenges, particularly due to the repeated solution of stiff ODE systems required during parameter estimation and optimization. Several strategies have been explored in prior work to accelerate these computations, including surrogate modeling of stiff ODEs and advection–dispersion equations [26–28], quantitative systems pharmacology (QSP) model emulation [29], continuous-time echo state networks (CTESN) [30], neural ODEs [31, 32], and hybrid mechanistic–ML frameworks [23, 33, 34]. These approaches suggest that ML-based surrogates can faithfully approximate ODE outputs while reducing computational cost.

Building on this body of work, the present study focuses on a large-scale empirical evaluation of surrogate modeling within a clinically grounded hybrid framework for RVTE prediction. Rather than introducing the surrogate concept, we investigate the application-specific performance of replacing a mechanistic blood coagulation model with ML surrogates that directly map patient-specific kinetic parameters to ETP. Therefore, it is necessary to rigorously assess the trade-offs among surrogate fidelity, computational throughput, and downstream diagnostic performance when embedded in a hybrid optimization pipeline to identify the best surrogate models and optimizer algorithms.

To this end, we present a comprehensive benchmark of 14 distinct ML algorithms acting as surrogates for a high-dimensional, stiff coagulation ODE system. We evaluate their integration with various metaheuristic optimizers under a unified experimental protocol. By jointly analyzing optimization trajectory behavior, surrogate approximation error, and final RVTE classification accuracy, this work provides evidence-based guidance for selecting surrogate–optimizer pairs. The primary contribution of this study lies in the RVTE-specific integration of surrogate modeling into a hybrid mechanistic–ML framework and in the breadth of the empirical evaluation. By systematically benchmarking surrogate–optimizer combinations, this work provides practical, application-driven guidance for designing and optimizing hybrid data-driven and mechanistic modeling pipelines for hematological diseases, particularly those involving blood clot formation and growth.

2 Materials and methods

Figure 1 summarizes the workflow pipeline used in this study. The main steps are: (i) generate an input matrix by varying the values of kinetic parameters of a blood coagulation cascade represented by a system of ODEs, (ii) compute the target variable (ETP), (iii) train 14 ML models to predict ETP from kinetic parameters, (iv) replace the ODE solver with surrogate models in a hybrid framework linking clinical and hematological data to kinetic parameters, ETP, and clinical outcome (RVTE or non-RVTE), (v) optimize the hybrid model using eight metaheuristic algorithms, and (vi) evaluate surrogate model performance in accuracy and optimization acceleration. Further details are provided in the following subsections.

2.1 Dataset generation

The dataset used for surrogate model training was generated from a mechanistic ODE system describing the blood coagulation cascade. The underlying model of the extrinsic pathway, initially proposed by Zhu [35], captures the dynamic interactions among 27 biochemical species via 41 kinetic reactions and has been used in previous studies [23, 36]. Baseline kinetic parameters, including catalytic rate constants, Michaelis-Menten constants, and other rate constants, along with species' initial concentrations and mathematical equations, are detailed in the Supplementary Material (SM).

A Sobol quasi-random sampling method was employed to generate 100,000 parameter sets by varying each baseline kinetic parameter between 50% and 300% of its nominal value, ensuring uniform coverage of the high-dimensional input space. The choice of sampling kinetic parameters within 50–300% of their nominal values was motivated by the recognition that coagulation kinetics exhibit substantial interindividual variability and are influenced by demographic, clinical, and experimental factors [23, 34]. Although many mechanistic coagulation models assume fixed kinetic constants, prior studies have demonstrated that these parameters can vary widely across experimental conditions and model formulations, with some kinetic parameters differing by orders of magnitude across published models [24]. Previous analyses have explored kinetic parameter ranges spanning 50–150% [37] and even 10–1000% [38] around nominal values. In this context, the selected 50–300% sampling range

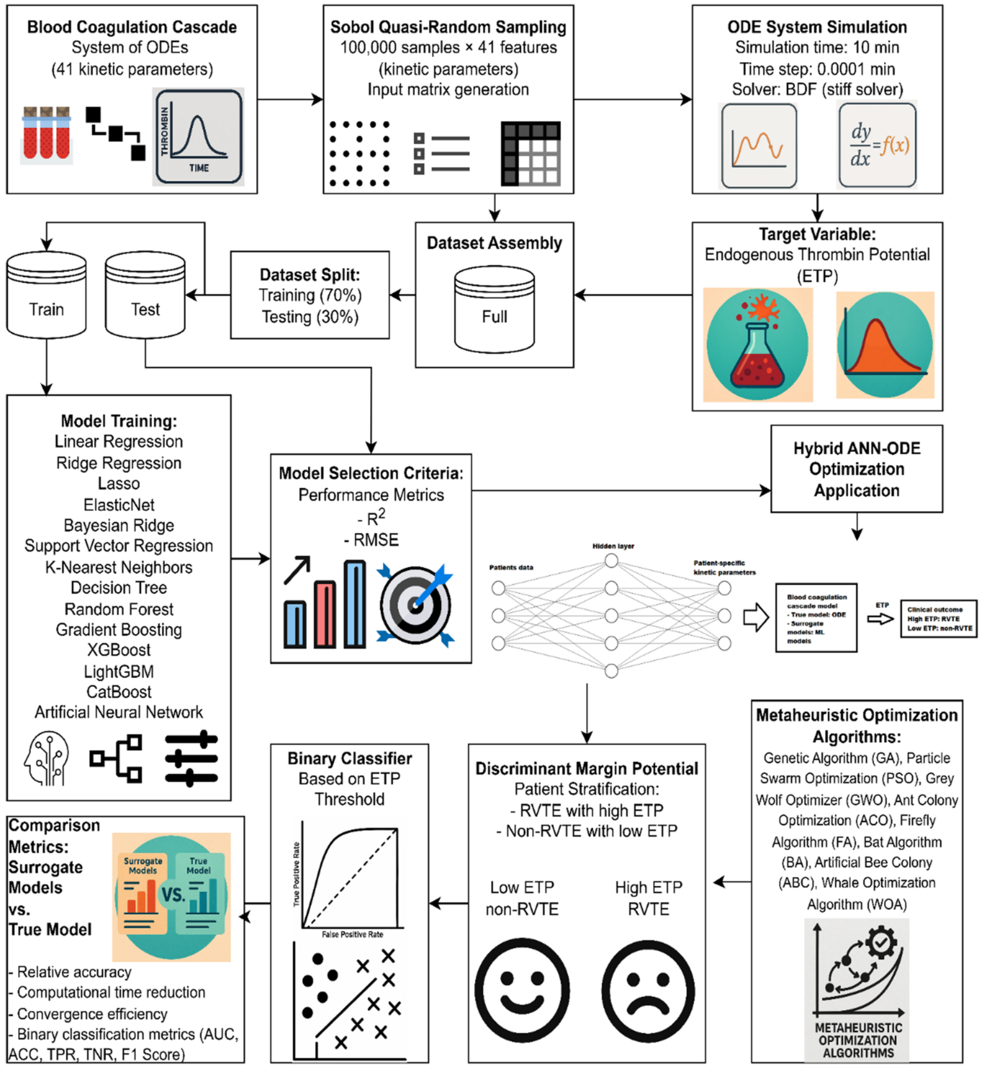

Fig. 1 Overview of the computational workflow for surrogate modeling and hybrid optimization

represents a defensible compromise that captures biologically plausible variability while maintaining numerical stability and interpretability.

Each sampled parameter set was used to numerically solve the ODE system over a 10-minute interval with a time step of 0.0001 min, using the backward differentiation formula (BDF) method for stiff systems. The output of interest, thrombin concentration, was integrated over time via the trapezoidal method to compute the ETP, a widely used measure of thrombin generation capacity.

The final dataset comprised 100,000 samples, each with 41 input features (kinetic parameters) and one target variable (ETP, computed via the BDF solver). The dataset was randomly split into training (70%) and testing (30%) subsets. To prevent data leakage, all preprocessing steps, including normalization, model training, and hyperparameter tuning, were conducted exclusively on the training set. The test set was reserved solely for evaluating model performance on unseen data.

2.2 Machine learning models training and evaluation

To develop surrogate models for predicting ETP, we evaluated 14 ML algorithms: Linear Regression (LR), Ridge Regression (RR), Lasso Regression (Lasso), ElasticNet Regression (EN), Bayesian Ridge Regression (BR), Support Vector Regression (SVR), K-Nearest Neighbors (KNN), Decision Tree (DT), Random Forest (RF), Gradient

Boosting (GB), eXtreme Gradient Boosting (XGBoost), Light Gradient Boosting Machine (LightGBM), Categorical Boosting (CatBoost), and Artificial Neural Network (ANN). Input features were normalized using Z-score normalization, computed as $z = (x - \mu)/\sigma$, where μ and σ are the mean and standard deviation of the training data, respectively. The target variable (ETP) was also normalized using the same Z-score method to improve numerical stability and model convergence.

Each surrogate model was trained using normalized input features derived exclusively from the training set. Model development employed 5-fold cross-validation within the training data, and performance metrics were computed on the corresponding validation folds to guide hyperparameter selection. This procedure ensured robust model tuning while preventing information leakage from the independent test set.

After hyperparameter optimization, the final model configuration for each algorithm was retrained on the full training set and subsequently evaluated on the held-out test set, which was not used at any stage during training or model selection. Model predictions were inverse-transformed back to the original ETP scale prior to performance evaluation, ensuring that all reported error metrics were expressed in clinically interpretable units (nM·min).

Predictive performance was quantified using the root mean squared error (RMSE) and the coefficient of determination (R^2), defined in Eqs. (1) and (2), respectively. Those metrics were calculated for both training and testing sets.

For each algorithm, a comprehensive grid-based hyperparameter search was performed over a predefined parameter space. All possible combinations of hyperparameter values were systematically evaluated, and each candidate model was trained and validated using the cross-validation framework described above. The complete hyperparameter grids explored for each algorithm are reported in Table 1.

Figure 2 provides a schematic representation of the experimental workflow, highlighting the strict separation between training and testing partitions to prevent data leakage during feature scaling and hyperparameter optimization. This pipeline ensures that the independent test set remains isolated until the final performance assessment.

All computations were implemented in MATLAB version R2021, and Python v3.12.7, using the following libraries and versions: NumPy v1.26.4, Pandas v2.2.2, scikit-learn v1.5.1, XGBoost v2.1.2, LightGBM v4.5.0, CatBoost v1.2.7, and joblib v1.4.2. These libraries were used for data preprocessing, model implementation, training, and evaluation. All computational tasks were executed on a system with an AMD Ryzen 7900X processor, an NVIDIA RTX 4090 GPU, and 32 GB of DDR5 RAM, running Windows 11. When feasible, parallel computing was used.

$$RMSE = \sqrt{\frac{1}{n}\sum_{i=1}^{n}\left(y_i - \widehat{y}_i\right)^2} \tag{1}$$

$$R^2 = 1 - \frac{\sum_{i=1}^{n}\left(y_i - \widehat{y}_i\right)^2}{\sum_{i=1}^{n}\left(y_i - \bar{y}\right)^2} \tag{2}$$

Here, $RMSE$ is the RMSE, R^2 is the R^2, n is the number of data points, y_i is the actual value, $\widehat{y}_i$ is the predicted value, and $\bar{y}$ is the mean of the actual values.

2.3 Hybrid ANN-surrogate modeling

The model used in this study to relate patient clinical data, kinetic parameters, ETP, and clinical outcome (RVTE) was based on the framework established in our previous study [23]. This study utilized clinical and hematological data from 235 patients who were monitored following a confirmed VTE event. Among these, 49 patients experienced a RVTE episode after discontinuing anticoagulant therapy. Details of the clinical variables are available in prior publications [23, 36, 39]. For model development, stratified patient data splitting was applied, allocating 70% of the patients to the training set and the remaining 30% to the test set.

Table 1 Hyperparameter search spaces for grid search optimization of machine learning regression models

Model	Hyperparameters	Search space
Linear Regression (LR)	fit_intercept, positive	fit_intercept: [True, False]; positive: [True, False]
Ridge Regression (RR)	alpha, solver, fit_intercept	alpha: logspace(-4, 3, 100); solver: ['auto', 'svd', 'cholesky', 'lsqr', 'sparse_cg', 'sag', 'saga']; fit_intercept: [True, False]
Lasso Regression (Lasso)	alpha, max_iter, fit_intercept, selection	alpha: concat of four linspace ranges (0.0001–10); max_iter: [10000, 20000, 30000]; fit_intercept: [True, False]; selection: ['cyclic', 'random']
ElasticNet Regression (EN)	alpha, l1_ratio, max_iter, fit_intercept	alpha: concat of four linspace ranges (0.0001–10); l1_ratio: linspace(0, 1, 11); max_iter: [10000, 20000, 30000]; fit_intercept: [True, False]
Bayesian Ridge Regression (BR)	n_iter, alpha_1, alpha_2, lambda_1, lambda_2	n_iter: [100, 200, 300, 400, 500]; alpha_1: [1e-6, 1e-5, 1e-4, 1e-3]; alpha_2: [1e-6, 1e-5, 1e-4, 1e-3]; lambda_1: [1e-6, 1e-5, 1e-4, 1e-3]; lambda_2: [1e-6, 1e-5, 1e-4, 1e-3]
Support Vector Regression (SVR)	C, epsilon, kernel, gamma, degree	C: logspace(-3, 3, 25); epsilon: linspace(0.001, 0.5, 25); kernel: ['linear', 'rbf', 'poly']; gamma: logspace(-4, 0, 20) — for 'rbf' and 'poly'; degree: [2, 3, 4, 5] — only if kernel='poly'
K-Nearest Neighbors (KNN)	n_neighbors, weights, p, algorithm, leaf_size, metric, metric_params	n_neighbors: [1–50]; weights: ['uniform', 'distance']; p: [1, 2, 3]; algorithm: ['auto', 'ball_tree', 'kd_tree', 'brute']; leaf_size: [10, 15, 20, 25, 30, 35, 40, 45, 50]; metric: ['minkowski', 'euclidean', 'manhattan', 'chebyshev']; metric_params: [None]
Decision Tree (DT)	max_depth, min_samples_split, min_samples_leaf, max_features	max_depth: [1–30]; min_samples_split: [2–20]; min_samples_leaf: [1–20]; max_features: [None, 'auto', 'sqrt', 'log2']
Random Forest (RF)	n_estimators, max_depth, min_samples_split, min_samples_leaf, max_features, bootstrap	n_estimators: [50, 100, 150, 200, 250, 300]; max_depth: [None, 5, 10, 15, 20, 25, 30]; min_samples_split: [2–10]; min_samples_leaf: [1–10]; max_features: [None, 'sqrt', 'log2']; bootstrap: [True, False]
Gradient Boosting (GB)	n_estimators, learning_rate, max_depth, min_samples_split, min_samples_leaf, max_features, subsample	n_estimators: [50, 100, 150, 200, 250, 300]; learning_rate: [0.01, 0.05, 0.1, 0.2]; max_depth: [3–15]; min_samples_split: [2–10]; min_samples_leaf: [1–10]; max_features: [None, 'sqrt', 'log2']; subsample: [0.5, 0.7, 0.8, 1.0]
eXtreme Gradient Boosting (XGBoost)	n_estimators, learning_rate, max_depth, subsample, colsample_bytree, gamma, reg_alpha, reg_lambda	n_estimators: [50, 100, 150, 200, 250, 300]; learning_rate: [0.01, 0.05, 0.1, 0.2]; max_depth: [3–10]; subsample: [0.5, 0.7, 0.8, 1.0]; colsample_bytree: [0.5, 0.7, 0.8, 1.0]; gamma: [0, 0.1, 0.2, 0.3]; reg_alpha: [0, 0.01, 0.1, 1]; reg_lambda: [0.5, 1, 1.5, 2]
Light Gradient Boosting Machine (LightGBM)	n_estimators, learning_rate, max_depth, num_leaves, subsample, colsample_bytree, reg_alpha, reg_lambda	n_estimators: [50, 100, 150, 200, 250, 300]; learning_rate: [0.01, 0.05, 0.1, 0.2]; max_depth: [-1, 0, 1, 2, …, 10] (-1 = no limit); num_leaves: [20, 30, 40, …, 150]; subsample: [0.5, 0.7, 0.8, 1.0]; colsample_bytree: [0.5, 0.7, 0.8, 1.0]; reg_alpha: [0, 0.01, 0.1, 1]; reg_lambda: [0.5, 1, 1.5, 2]
Categorical Boosting (CatBoost)	iterations, learning_rate, depth, l2_leaf_reg, bagging_temperature, border_count	iterations: [50, 100, 150, 200, 250, 300]; learning_rate: [0.01, 0.05, 0.1, 0.2]; depth: [3–10]; l2_leaf_reg: [1, 3, 5, 7, 9]; bagging_temperature: [0, 0.5, 1, 2]; border_count: [32, 64, 128, 254]
Artificial Neural Network (ANN)	hidden_layers, neurons_per_layer, activation, solver	hidden_layers: [1, 2]; neurons_per_layer: [2–30]; activation: ['tansig', 'logsig', 'purelin', 'satlins']; solver: 'Levenberg-Marquardt'

In this study, we aimed to replace the computationally intensive ODE system with surrogate models, as illustrated in Fig. 3. To achieve this, various combinations of model configurations and hyperparameters were explored, as summarized in Table 2. The primary goal was to optimize the weights and biases of the ANN, which maps patient-specific clinical and hematological data to kinetic parameters, as formalized in Eq. 3.

$$\boldsymbol{K} = satlins\left[\boldsymbol{W}_2 \cdot tansig\left(\boldsymbol{W}_1 \cdot \boldsymbol{X} + \boldsymbol{b}_1\right) + \boldsymbol{b}_2\right] \quad (3)$$

In this formulation, $\boldsymbol{W}_1 \in \mathbb{R}^{n_n \times n_f}$ and $\boldsymbol{W}_2 \in \mathbb{R}^{n_k \times n_n}$ are the weight matrices of the first and second layers, respectively. $\boldsymbol{b}_1 \in \mathbb{R}^{n_n}$ and $\boldsymbol{b}_2 \in \mathbb{R}^{n_k}$ are the corresponding bias vectors. The input feature vector $\boldsymbol{X}$ passes through a two-layer feedforward ANN, where the hidden layer uses, in this study, a hyperbolic tangent sigmoid activation function (*tansig*), and the output layer applies a symmetric saturating linear function (*satlins*) to constrain values to the range $[-1, 1]$. The network output is subsequently linearly scaled to represent kinetic parameters within 50% to 300% of their nominal baseline values, resulting in the final transformed output $\boldsymbol{K_r} \in \mathbb{R}^{n_k \times n_o}$. Here, n_f, n_o, n_n, and n_k correspond to the numbers of input features, observations, hidden-layer neurons, and kinetic parameters, respectively.

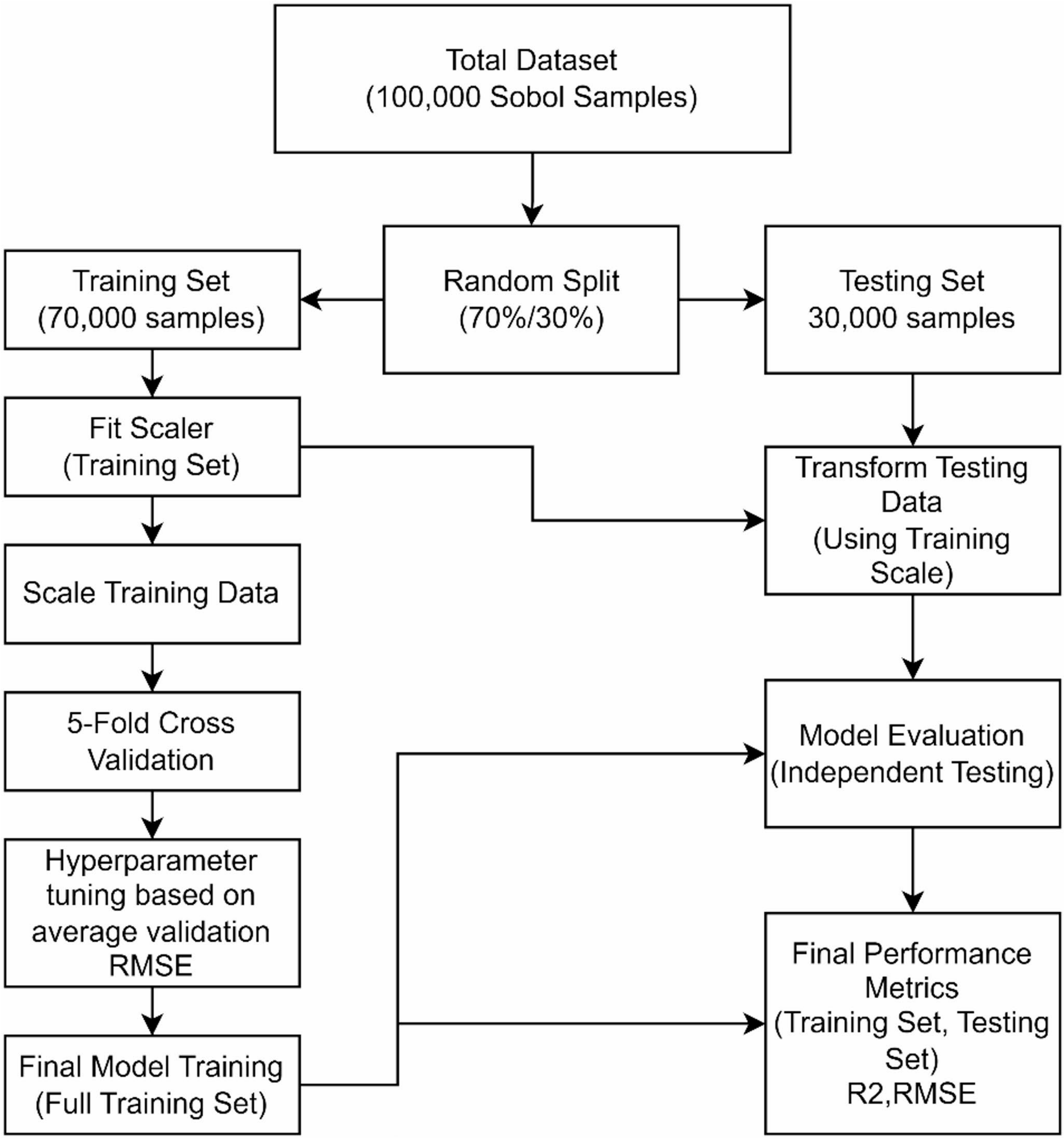

Fig. 2 Schematic of the data partitioning, preprocessing, and model optimization pipeline of surrogate models

The predicted kinetic parameters for each patient ($\boldsymbol{K_r}$) were then input into the trained surrogate models to estimate ETP values. These ETP predictions were used to classify patients as RVTE-positive (ETP above the threshold) or non-RVTE (ETP below the threshold). The ETP threshold was determined by maximizing Youden's J index on the receiver operating characteristic (ROC) curve. In addition to ROC analysis, several standard classification metrics were calculated to assess model performance, including area under the curve (AUC), accuracy (ACC), true positive rate (TPR), true negative rate (TNR), and F1-Score, as defined in Eqs. 4–7, respectively.

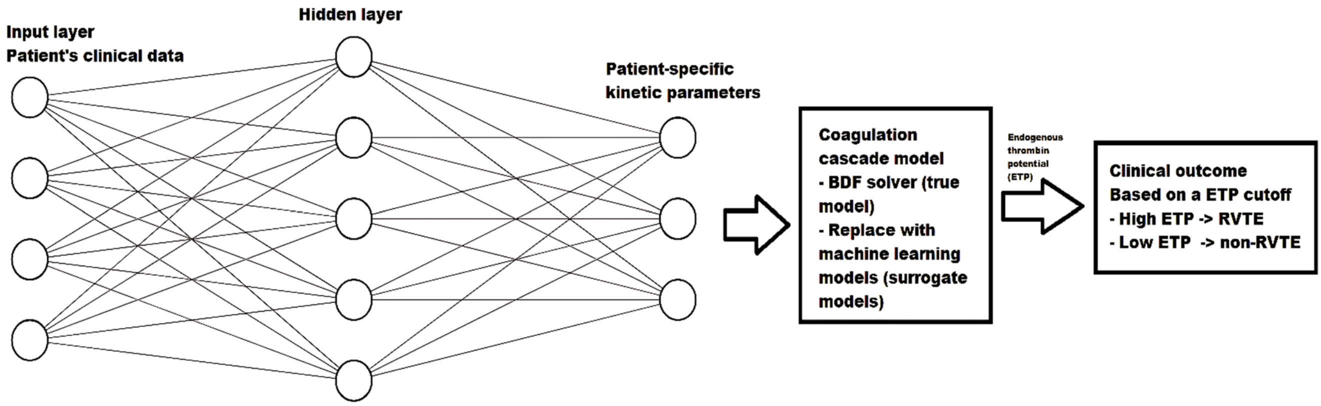

Fig. 3 Hybrid ANN-Surrogate modeling framework for predicting RVTE risk based on patient-specific clinical data and coagulation kinetics

Table 2 Summary of model configuration options explored for hybrid ANN-Surrogate framework optimization

Configuration option	Description	Evaluated settings
Patient input data normalization scheme	Technique applied to preprocess and scale input variables.	- *MinMax*: Rescales inputs to the interval [−1, 1]. - *Standard*: Centers data to zero mean and unit variance (Z-score).
Dimensionality reduction strategy	Determines whether all features or a reduced set are used as model inputs.	- *All Features*: Incorporates the complete clinical dataset [39]. - *PCA-Based*: Utilizes Principal Component Analysis to reduce dimensionality [39].
Model target scope	Defines the extent of kinetic parameters that the model is trained to predict.	- *Comprehensive*: Outputs the complete set of kinetic parameters. - *Selective*: Outputs a subset identified as most influential through sensitivity analysis [23].
Neural network width	Specifies the number of units in the hidden layer, impacting model flexibility.	{20, 30, 40, 50}
Initial state specification	Approach for Assigning Starting Biochemical Conditions in Simulations.	- *Uniform*: All subjects begin with the same predefined initial values [35, 36]. - *Personalized*: Initial conditions are tailored to individual patient data [36].
Optimization criterion	Maximize an objective function using metaheuristic optimization algorithms.	- *Margin-Only*: Uses Discriminant Margin Potential (DMP) alone. - *Sparse*: Combines DMP with L1 regularization. - *Full Regularization*: Combines DMP with both L1 and L2 penalties.

$$ACC = \frac{TP + TN}{TP + TN + FP + FN} \tag{4}$$

$$TPR = \frac{TP}{TP + FN} \tag{5}$$

$$TNR = \frac{TN}{TN + FP} \tag{6}$$

$$F1Score = \frac{2TP}{2TP + FP + FN} \tag{7}$$

Here, TP is the true positives, TN is the true negatives, FP is the false positives, and FN is the false negatives.

2.4 Optimization

The primary objective is to optimize the neural network's weights and biases (Eq. 3) such that patients with RVTE exhibit higher predicted ETP values, while those without RVTE exhibit lower values. To achieve this, the training process aims to maximize the discriminant margin potential (DMP), which quantifies the separation between the two patient groups based on their ETP values. This function is formally defined in Eq. 8.

$$DMP = \left|\sum_{i\in\mathcal{P}} \boldsymbol{Y}_{\boldsymbol{output},i}\right| \bullet \frac{n_n}{max\,(1, n_p)} - \left|\sum_{i\in\mathcal{N}} \boldsymbol{Y}_{\boldsymbol{output},i}\right| \tag{8}$$

Here, $\boldsymbol{Y}_{\boldsymbol{output}} \in \mathbb{R}^{n_o}$ represent the vector of predicted ETPs for all patients, and $\boldsymbol{Y}_{\boldsymbol{target}} \in \{-1, +1\}^{n_o}$ denote the corresponding binary class labels, where $+1$ indicates an RVTE-positive patient and -1 indicates a non-RVTE patient. Define the index sets of positive and negative samples as $\mathcal{P} = \{i|\boldsymbol{Y}_{\boldsymbol{target},\boldsymbol{i}} = +1\}$ and $\mathcal{N} = \{i|\boldsymbol{Y}_{\boldsymbol{target},\boldsymbol{i}} = -1\}$, respectively, with cardinalities $n_p = |\mathcal{P}|$ and $n_n = |\mathcal{N}|$. In Eq. 8, the first term reflects the aggregated predicted thrombin potential among RVTE-positive patients, weighted by the ratio of class sizes to mitigate class imbalance. The second term represents the corresponding aggregation for non-RVTE patients. The objective is to maximize this margin, thereby enhancing the model's discriminative capability between the two clinical outcomes.

To control model complexity and improve generalization, two regularized variants of the objective function were also considered. Let $\boldsymbol{W}_1 \in \mathbb{R}^{n_n \times n_f}$ and $\boldsymbol{W}_2 \in \mathbb{R}^{n_k \times n_n}$ denote the weight matrices of the hidden and output layers, respectively. The three objective functions to be maximized in the pipeline are:

$$F_{DMP} = DMP \tag{9}$$

$$F_{DMP,L1} = DMP - \frac{1}{n_o}\left[\sum(|\boldsymbol{W}_1|) + \sum(|\boldsymbol{W}_2|)\right] \tag{10}$$

$$\begin{aligned} F_{DMP,L1+L2} = DMP &- \frac{1}{n_p + n_n}\left[\sum(|\boldsymbol{W}_1|) + \sum(|\boldsymbol{W}_2|)\right] \\ &- \frac{1}{n_p + n_n}\left[\sum(\boldsymbol{W}_1 \odot \boldsymbol{W}_1) + \sum(\boldsymbol{W}_2 \odot \boldsymbol{W}_2)\right] \end{aligned} \tag{11}$$

Here, $\|\cdot\|_1$ and $\|\cdot\|_2$ denote denote the ℓ_1 and ℓ_2 norms, respectively. The ℓ_1 penalty promotes sparsity in network weights, while the combined $\ell_1 + \ell_2$ formulation balances sparsity with smoothness and weight shrinkage.

Because the DMP-based objectives are non-differentiable due to the model's underlying ODE system, gradient-based optimization is not applicable. Consequently, all objective functions were maximized using population-based metaheuristic optimization algorithms (MOAs), as detailed in Sect. 2.6.

This study used eight MOAs: Genetic algorithm (GA) [40], particle swarm optimization (PSO) [41], grey wolf optimizer (GWO) [42], ant colony optimizer (ACO) [43], firefly algorithm (FA) [44], bat algorithm (BA) [45], artificial bee colony (ABC) [46], and whale optimizer algorithm (WOA) [47]. Additional implementation details, including pseudocode and hyperparameter configurations for each algorithm, are provided in the SM. In all cases, 100 individuals and 300 generations were used.

2.5 Metrics for performance evaluation of surrogate models

The substitution of the ODE system with ML-based surrogate models within the hybrid framework was evaluated by comparing their outputs against reference values obtained using the original ANN–ODE approach, using the same optimized adjustable parameters, which involved numerical integration of the ODEs via the BDF method with a time step of 0.0001 min over a 10-minute simulation horizon. To assess the accuracy of the surrogate models, several performance metrics were computed, as summarized in Table 3. A radar plot was also constructed to facilitate the visualization of the performance across ML models and MOAs.

2.6 Improved analysis on the best surrogate model

After identifying the best-performing ML surrogate to replace the ODE system, additional analyses were conducted to better characterize its optimization behavior and robustness across the full framework.

Table 3 Quantitative metrics for assessing accuracy and efficiency of surrogate models within the hybrid ANN–ODE framework

Type	Metric	Mathematical Equation	Notation	Equation #
Normalized per ML	ML-normalized true performance (%)	$\tilde{P}_{i,j}^{ML} = \frac{P_{i,j}}{\max_{j} P_{i,j}} \times 100$	$P_{i,j}$: Performance of ML model with MOAj; max across MOAs for fixed ML	(12)
Normalized per MOA	MOA-normalized true performance (%)	$\tilde{P}_{i,j}^{MOA} = \frac{P_{i,j}}{\max_{i} P_{i,j}} \times 100$	$P_{i,j}$: Performance of ML model with MOAj; max across MLs for fixed MOA	(13)
–	Relative accuracy (%)	$RA_{i,j} = 100 - \left\lvert \frac{DMP_{i,j}^{sur} - DMP_{i,j}^{true}}{DMP_{i,j}^{true}} \right\rvert \times 100$	Relative Accuracy (RA) measures the percentage agreement between the Discriminant Margin Potential (DMP) from the surrogate model and the full ODE-based model for the -th ML model and$j$-th MOA. Higher values indicate closer alignment.	(14)
Normalized per MOA	MOA-normalized relative accuracy (%)	$\tilde{RA}_{i,j}^{MOA} = \frac{RA_{i,j}}{\max_{i} RA_{i,j}} \times 100$	Normalized across MLs for each MOA	(15)
–	Model time reduction (%)	$TR_{i,j} = \left(1 - \frac{T_{i,j}^{sur}}{T_{j}^{true}}\right) \times 100$	$T_{i,j}^{sur}$: Computational time with surrogate of ML model with MOAj.T_{j}^{true}: Computational time for full ODE on MOAj	(16)
Normalized per MOA	MOA-normalized time reduction (%)	$\tilde{TR}_{i,j}^{MOA} = \frac{TR_{i,j}}{\max_{i} TR_{i,j}} \times 100$	Normalized across MLs for each MOA	(17)
–	Convergence efficiency (%)	$CE_{i,j} = \left(1 - \frac{G_{i,j}^{90}}{G_{i,j}^{total}}\right) \times 100$	$G_{i,j}^{90}$: generation reaching 90% of the best fitness.$G_{i,j}^{total}$: total generations	(18)
–	Relative accuracy AUC (%)	$RA_{i,j}^{AUC} = 100 - \left\lvert \frac{AUC_{i,j}^{sur} - AUC_{i,j}^{true}}{AUC_{i,j}^{true}} \right\rvert \times 100$	Measures the percentage agreement between the Area Under the Curve (AUC) predicted by the surrogate model and the full ODE model for the i-th ML model and j-th MOA. Higher values indicate closer alignment.	(19)
–	Relative accuracy ACC (%)	$RA_{i,j}^{ACC} = 100 - \left\lvert \frac{ACC_{i,j}^{sur} - ACC_{i,j}^{true}}{ACC_{i,j}^{true}} \right\rvert \times 100$	Quantifies agreement between surrogate and true accuracy (ACC). Values closer to 100% reflect better surrogate fidelity	(20)
–	Relative accuracy TPR (%)	$RA_{i,j}^{TPR} = 100 - \left\lvert \frac{TPR_{i,j}^{sur} - TPR_{i,j}^{true}}{TPR_{i,j}^{true}} \right\rvert \times 100$	Represents the agreement in true-positive rates between the surrogate and full models. Higher scores denote better match.	(21)
–	Relative accuracy TNR (%)	$RA_{i,j}^{TNR} = 100 - \left\lvert \frac{TNR_{i,j}^{sur} - TNR_{i,j}^{true}}{TNR_{i,j}^{true}} \right\rvert \times 100$	Assesses the agreement in true negative rates between surrogate and full ODE predictions. Higher percentages indicate stronger concordance.	(22)
–	Relative accuracy F1-score (%)	$RA_{i,j}^{F1} = 100 - \left\lvert \frac{F1_{i,j}^{sur} - F1_{i,j}^{true}}{F1_{i,j}^{true}} \right\rvert \times 100$	Evaluates the similarity of F1-scores from surrogate and true models, with higher values reflecting better accuracy.	(23)
Normalized per MOA	MOA-normalized RA AUC (%)	$\tilde{RA}_{i,j}^{AUC} = \frac{RA_{i,j}^{AUC}}{\max_{i} RA_{i,j}^{AUC}} \times 100$	Best AUC-aligned ML model per MOA scores 100%.	(24)
Normalized per MOA	MOA-normalized RA ACC (%)	$\tilde{RA}_{i,j}^{ACC} = \frac{RA_{i,j}^{ACC}}{\max_{i} RA_{i,j}^{ACC}} \times 100$	Normalized accuracy agreement.	(25)
Normalized per MOA	MOA-normalized RA TPR (%)	$\tilde{RA}_{i,j}^{TPR} = \frac{RA_{i,j}^{TPR}}{\max_{i} RA_{i,j}^{TPR}} \times 100$	Normalized TPR agreement.	(26)
Normalized per MOA	MOA-normalized RA TNR (%)	$\tilde{RA}_{i,j}^{TNR} = \frac{RA_{i,j}^{TNR}}{\max_{i} RA_{i,j}^{TNR}} \times 100$	Normalized TNR agreement.	(27)
Normalized per MOA	MOA-normalized RA F1-score (%)	$\tilde{RA}_{i,j}^{F1} = \frac{RA_{i,j}^{F1}}{\max_{i} RA_{i,j}^{F1}} \times 100$	Normalized F1-score agreement.	(28)

2.6.1 Comparison with gradient-based optimizer algorithms

Once a differentiable surrogate replaced the ODE system, the optimization pipeline became end-to-end differentiable. Therefore, in addition to the previously employed metaheuristic algorithms, three widely used gradient-based optimizers were implemented as baselines: stochastic gradient descent (SGD) [48], root mean square propagation (RMSProp) [49], and adaptive moment estimation (Adam) [50]. All optimizers were tasked with maximizing the same fitness function used in the metaheuristic experiments, thereby ensuring a fair comparison.

Adam optimization was implemented with a learning rate of 10^{-3}, first- and second-moment decay parameters ${}_1=0.9$ and ${}_2=0.999$, and numerical stabilization constant $=10^{-8}$. RMSProp used the same learning rate, a decay factor of 0.9 for the running average of squared gradients, and $=10^{-8}$. SGD was implemented with a higher learning rate and Xavier initialization to mitigate vanishing gradients in early iterations. In preliminary analyses, gradient-based optimizers exhibited substantially lower per-iteration computational cost, with average runtimes approximately three times faster than those of MOAs. To ensure a fair comparison under comparable computational budgets, all gradient-based optimizers were therefore executed for a fixed number of 900 iterations, corresponding to three times the iteration count used for MOAs, resulting in similar overall wall-clock optimization times across methods.

Gradient-based optimizers were compared with MOAs in terms of convergence speed, optimization stability, computational time, and predictive performance.

2.6.2 Improvements in the binary classifier

To quantify the impact of replacing the mechanistic ODE–based objective with a surrogate model on downstream RVTE prediction, we conducted a systematic comparative evaluation using the top-performing ML surrogate and the original ODE simulator.

To quantify predictive uncertainty, non-parametric bootstrap resampling (10,000 iterations) was applied to both training and test sets independently. For each resample, standard classification metrics were computed, including the AUC, ACC, TPR, TNR, and F1-score. Final estimates are reported as bootstrap means with 95% confidence intervals (CIs) defined by the 2.5th and 97.5th percentiles. Differences in performance between surrogate-driven and ODE-driven optimization pipelines were interpreted as a direct measure of surrogate-induced uncertainty propagation to the RVTE classification task.

To formally assess differences in discriminative performance across optimization strategies, pairwise comparisons between GWO and alternative optimizers were performed using DeLong's test, which accounts for the correlation between models evaluated on the same patient samples. Mean AUC differences (ΔAUC), 95% confidence intervals, and two-sided p-values were reported, with statistical significance defined at $\alpha=0.05$.

Model performance was further evaluated using complementary calibration, clinical utility, and threshold analyses. Ground-truth outcomes were defined by the binary variable ${}_{\text{true}}$. For calibration and decision curve analyses, continuous model outputs were min–max normalized to the [0,1] interval to represent predicted RVTE risk. In contrast, threshold analyses were conducted exclusively on the original ETP values (nM·min) to preserve clinical interpretability.

Calibration was assessed using quantile-based binning with five equally populated bins. For each bin, the mean predicted risk was compared with the observed event rate. Uncertainty in calibration estimates was quantified via non-parametric bootstrapping (10,000 resamples), and 95% CIs were constructed for observed proportions. The 45° identity line represented perfect calibration.

Clinical usefulness was evaluated using decision curve analysis (DCA). The net benefit was computed across threshold probabilities ranging from 0.05 to 0.60 and compared with default strategies of treating all patients or none. This analysis quantified the trade-off between true positives and false positives while explicitly incorporating the relative harm of unnecessary treatment.

Finally, decision thresholds were explored using Youden's index (sensitivity + specificity − 1) evaluated over a dense grid of candidate ETP thresholds. Rather than selecting the single maximum Youden value, percentile-based criteria (85th–99th percentiles, model-specific) were applied to favor robust and conservative threshold selection. For each model, the lowest ETP threshold achieving the selected percentile was retained, and the corresponding sensitivity and specificity were reported.

2.6.3 Impact of pipeline design choices on RVTE prediction performance

To quantify the influence of pipeline configuration choices on RVTE prediction performance, we conducted a structured sensitivity analysis focusing on test-set discrimination, measured by the AUC. For each metaheuristic optimizer, results from all evaluated configurations were aggregated into a single dataset comprising framework-level settings, including data normalization strategy, initial condition model, feature selection approach, kinetic parameter subset, fitness score definition, and network size (number of hidden neurons).

A multivariable linear model was fitted with test-set AUC as the dependent variable, and all framework settings entered simultaneously as predictors. Categorical settings were modeled as factors, while the number of hidden neurons was treated as a continuous variable. Statistical significance of each setting was assessed using analysis of variance (ANOVA), with a significance threshold of $p < 0.05$.

To quantify the relative importance of each framework component, effect sizes were estimated using a drop-one partial R^2 approach. For each setting, a reduced model excluding that factor was fitted, and partial R^2 was computed as the proportion of variance explained by the omitted factor relative to the unexplained variance. This metric reflects the contribution of each pipeline component to predictive performance independently of other settings.

For statistically significant factors, the direction and magnitude of their impact were further characterized by computing the mean test-set AUC across all levels of the setting. The best-performing configuration was identified as the level with the highest mean AUC, and performance dispersion was summarized using ΔAUC, the difference between the highest and lowest mean AUC across levels. Only statistically significant framework components were reported in the final analysis.

2.6.4 Ablations on clinical input features and kinetic parameter subsets

To investigate the relationship between clinical input variables and learned kinetic parameters, Spearman rank correlation coefficients () were calculated using pairwise complete observations. This approach allowed for the quantification of monotonic associations while minimizing the influence of potential outliers.

Initial associations were estimated across the full patient cohort to establish a global association landscape. Statistical uncertainty and significance were assessed using a non-parametric bootstrap framework with 10,000 iterations. For each iteration, samples of the same size as the original dataset were drawn with replacement. From the resulting bootstrap distribution, percentile-based 95% CIs were derived. An association was deemed statistically significant if its 95% CI did not encompass zero.

To evaluate the impact of clinical outcomes on these associations, a stratified analysis was performed based on RVTE status (RVTE− and RVTE+). Correlation coefficients and 95% CIs were estimated independently within each stratum using the resampling strategy described above, with bootstrap sampling restricted to individuals within the respective subgroup.

Differential correlation analysis was subsequently performed to identify shifts in the association structure between groups. The metric Δ was defined as the absolute difference between the subgroup-specific Spearman coefficients (${}_{RVTE+} - {}_{RVTE-}$). To determine the statistical significance of these differences, 95% CIs for Δ were generated by independently resampling observations from both subgroups at each bootstrap iteration. A Δ was considered significant if the resulting interval excluded zero, indicating a statistically robust shift in the relationship between clinical inputs and kinetic parameters. Correlation matrices were visualized using heatmaps, with statistically significant associations indicated by overlaid black-filled circle markers.

2.6.5 Permutation-importance analysis

To quantify the relative contribution and global sensitivity of individual input features to the prediction of kinetic parameters, a permutation-importance analysis was conducted. This analysis was specifically applied to the top-performing surrogate-MOA models identified in the previous evaluation stages.

The permutation procedure involved independently shuffling each input feature within each sample while keeping all other variables at their baseline values. This process disrupts the specific association between a given feature and the output without altering its marginal distribution. For each feature, 500 independent permutations were performed to ensure statistical stability. Following each shuffling event, the ANN forward pass was recomputed, and the predicted kinetic parameters were compared against the original baseline predictions. Feature impact was quantified as the mean absolute deviation (MAD) between the permuted and baseline kinetic parameters, averaged across the entire dataset.

To facilitate comparison across kinetic parameters with varying absolute scales, importance values were normalized within each parameter. This was achieved by dividing each importance score by the maximum importance value observed across all input features for that specific parameter. The resulting normalized permutation-importance matrix represents the relative influence of each clinical input on the prediction of each kinetic parameter, where a value of 1.0 denotes the most influential feature. Finally, the importance landscape was visualized using heatmaps to identify global sensitivities and parameter-specific dependencies within the learned surrogate models.

2.6.6 Performance variability across independent runs

To account for the stochastic variability inherent to the framework, each algorithm was executed over 30 independent optimization runs using the best-performing configuration identified during the preliminary screening phase. Across runs, all algorithmic settings were held constant, including population size, stopping criterion, fitness function, surrogate model, and data preprocessing pipeline. Runs differed only in their random initialization, implemented via distinct random seeds.

For each run, we recorded (i) the final DMP value, (ii) convergence efficiency, and (iii) classification performance. Convergence efficiency was defined as 100% minus the percentage of generations required to reach 90% of the final fitness attained in that run, providing a normalized measure of optimization speed independent of absolute fitness scale. Predictive performance was evaluated using bootstrap-resampled AUC, computed separately on training and test sets using 10,000 nonparametric bootstrap iterations.

To summarize robustness and stability across runs, worst-case, mean, best-case, and standard deviation statistics were computed for each metric and each algorithm. To formally assess statistically significant differences among optimization algorithms across repeated runs, nonparametric rank-based statistical tests were employed, as described below. This multiple-run evaluation framework enabled a comprehensive assessment of optimization consistency, convergence behavior, and generalization performance under stochastic variability.

To compare the performance of the optimization algorithms across multiple independent runs, the Friedman test was employed [51, 52]. This nonparametric statistical test is widely used in algorithm benchmarking and is particularly well suited for repeated-measures designs, in which the same set of algorithms is evaluated across identical and independent runs, problems, or datasets [53–55]. Based on the DMP values, algorithms were ranked within each independent run, with higher-performing algorithms assigned better ranks. The Friedman test was then used to determine whether statistically significant differences existed among the algorithms' average ranks across runs.

When the Friedman test indicated a significant overall effect, pairwise post-hoc comparisons were conducted using the Nemenyi test. The Nemenyi procedure evaluates all pairwise differences in average ranks while

Table 4 Performance metrics of machine learning regression models on training and testing sets

Model	R^2 (Train)	R^2 (Test)	RMSE (Train)	RMSE (Test)
Linear	0.873	0.872	109.549	110.278
Ridge	0.873	0.872	109.549	110.278
Lasso	0.873	0.872	109.558	110.243
ElasticNet	0.873	0.872	109.553	110.255
Bayesian Ridge	0.873	0.872	109.549	110.278
Support Vector Regression	0.864	0.863	113.135	113.984
K Nearest Neighbors	0.999	0.615	0.001	190.798
Decision Tree	0.999	0.998	6.989	12.180
Random Forest	0.999	0.999	3.104	6.144
Gradient Boosting	0.999	0.999	0.939	3.373
XGBoost	0.999	0.999	2.831	3.701
LightGBM	0.999	0.999	2.831	4.175
CatBoost	0.999	0.999	3.088	3.439
ANN	**0.999**	**0.999**	**0.286**	**0.264**

The best-performing model is highlighted in bold

controlling the family-wise error rate, making it appropriate for multiple-algorithm comparisons under a common experimental protocol. Two-sided p-values were reported, with statistical significance defined at $\alpha = 0.05$.

3 Results

3.1 Comparative performance benchmarking of machine learning surrogates

The Sobol sequence generated 100,000 parameter sets, which were simulated using the ODE–based coagulation model to compute ETP, expressed in nM·min. The resulting ETP distribution had a mean of 768.7 nM·min (95% CI: 766.8–770.6), a median of 673.6 nM·min, and an interquartile range of 411.9 nM·min (Q1: 525.7 nM·min, Q3: 937.6 nM·min). The minimum and maximum observed values were 394.0 nM·min and 2212.1 nM·min, respectively. The literature describes ETP values covering a wide physiological spectrum, generally ranging from 554 to 1952 nM·min, depending on demographic and clinical factors such as age and comorbidities [56–60]. The ETP values generated by our simulations fall within and extend this reported range, consistent with the physiological range reported in the literature.

Table 4 summarizes the performance of the evaluated regression models on both the training and testing sets. LR, RR, Lasso, EN, and BR exhibited similar performance, with RMSE around 110 on both datasets, indicating moderate predictive accuracy without overfitting. SVR achieved slightly lower performance (RMSE ≈ 114). KNN severely overfitted the training set and performed poorly on the test set. Tree-based ensemble methods (DT, RF, GB, XGBoost, LightGBM, CatBoost) and the ANN achieved very high predictive accuracy, with $R^2 > 0.998$ and very low RMSE values, maintaining excellent generalization to the test data. Among them, ANN attained the highest performance, with $R^2 = 0.999$ and RMSE < 0.3 on both sets. The top-performing ANN has a structure with two hidden layers, *tansig*-*logsig*-*purelin* activation functions, and 41-7-10-1 neurons.

To further characterize surrogate model behavior beyond global goodness-of-fit metrics, residual-based analyses and formal statistical tests were conducted under two complementary evaluation settings: (i) an independent held-out test set spanning the full Sobol-sampled parameter space (50–300% of nominal kinetic values), and (ii) a constrained Sobol-sampled parameter set covering 95–105% of the nominal kinetic values, representing a clinically plausible local neighborhood around the baseline coagulation cascade model.

For each surrogate, predictive accuracy was quantified using the RMSE and mean absolute error (MAE). Systematic prediction bias was assessed using the mean residual (mean error). Statistical evidence of bias was evaluated exclusively using non-parametric methods: a sign test, which assesses whether residuals are symmetrically

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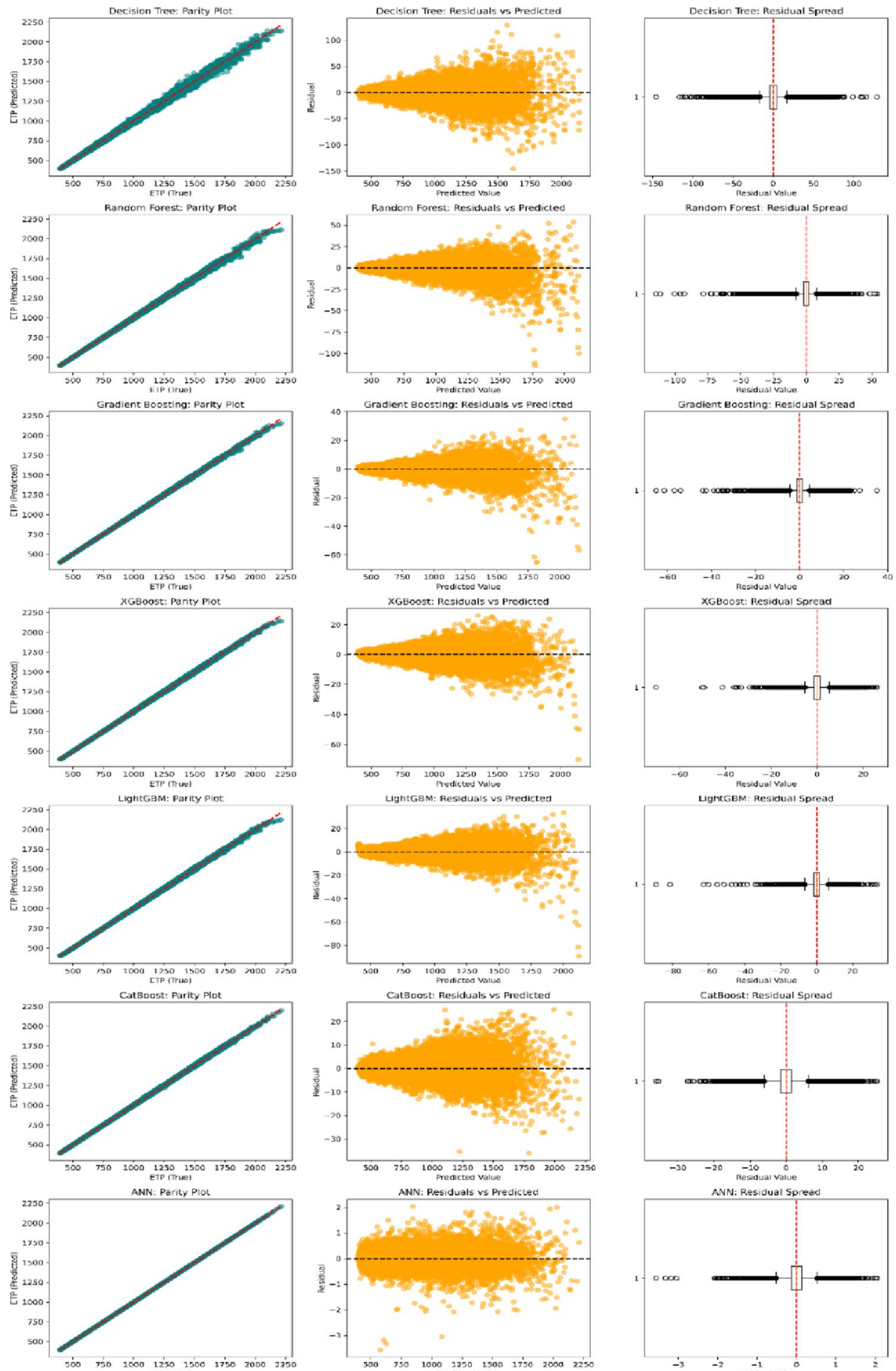

Fig. 4 Residual diagnostics of surrogate models on the independent test set sampled over the full Sobol parameter space (50–300% of nominal kinetic values)

distributed around zero, and the Wilcoxon signed-rank test, which additionally accounts for the magnitude of deviations while remaining robust to non-normal residual distributions.

Parity plots, residuals versus predicted values, and residual spread distributions are shown in Figs. 4 and 5 for the test and constrained sets, respectively. On the full test set (50–300%), DT and RF models exhibited the largest residual dispersion (DT MAE = 7.41; RF MAE = 3.52), with clear heteroscedasticity characterized by increasing

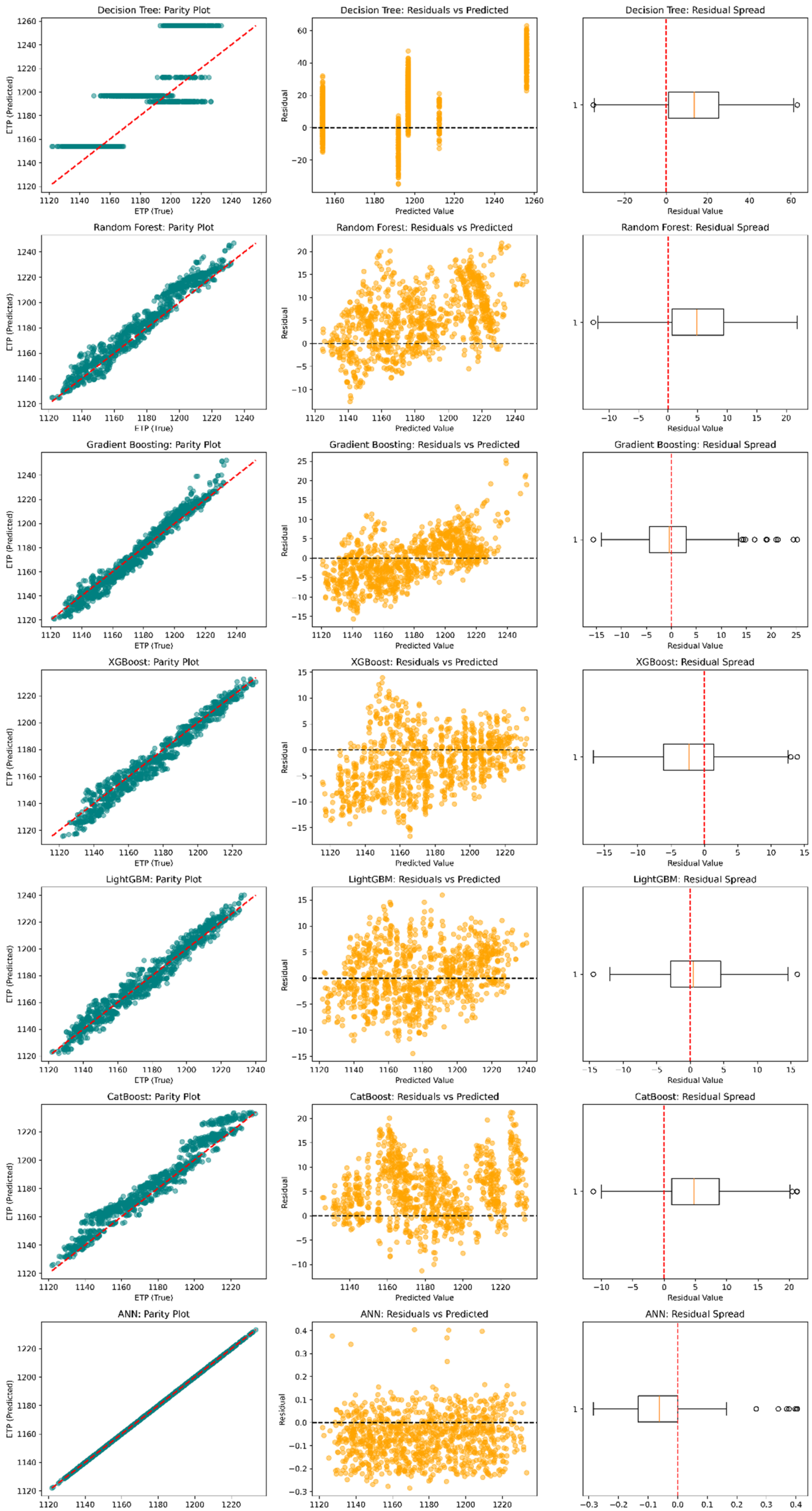

Fig. 5 Residual diagnostics of surrogate models evaluated on a constrained physiological parameter domain (95–105% of nominal kinetic values)

Table 5 Comparative analysis of surrogate-based optimization versus true ODE simulation using the decision tree model

MOA	Optimal DMP (Surrogate)	True DMP (Full ODE)	Relative Accuracy	Surrogate average Simulation Time (min)	Time Reduction (%)	Convergence efficiency
GA	145,873.39	129,043.06	86.96	5.59	99.41	53.00
PSO	225,716.37	209,133.48	92.07	5.58	99.70	65.00
GWO	228,392.24	250,110.55	91.32	5.51	99.41	45.00
ACO	80,218.55	78,394.15	97.67	5.80	99.39	78.00
FA	102,416.23	98,649.79	96.18	5.94	99.38	72.33
BA	199,976.94	210,642.69	94.94	5.80	99.39	68.67
ABC	164,248.32	181,892.29	90.30	12.40	99.35	67.00
WOA	102,416.23	58,021.55	23.46	5.28	99.45	99.33

Table 6 Comparative analysis of surrogate-based optimization versus true ODE simulation using the random forest model

MOA	Optimal DMP (Surrogate)	True DMP (Full ODE)	Relative Accuracy (%)	Surrogate average Simulation Time (min)	Time Reduction (%)	Convergence efficiency (%)
GA	123,081.79	102,212.49	79.58	327.78	65.48	58.67
PSO	237,265.31	253,091.02	93.75	323.89	82.64	40.33
GWO	238,781.78	273,844.50	87.20	325.06	65.39	47.02
ACO	86,848.91	86,217.74	99.27	322.65	65.80	31.33
FA	99,506.60	97,973.44	98.44	312.43	67.37	65.67
BA	198,671.81	213,758.18	92.94	316.02	66.52	37.67
ABC	156,879.52	162,294.17	96.66	640.82	66.24	55.33
WOA	87,780.41	75,764.70	85.91	320.60	66.85	92.67

Table 7 Comparative analysis of surrogate-based optimization versus true ODE simulation using the gradient boosting model

MOA	Optimal DMP (Surrogate)	True DMP (Full ODE)	Relative Accuracy (%)	Surrogate average Simulation Time (min)	Time Reduction (%)	Convergence efficiency (%)
GA	140,828.43	142,155.66	99.07	8.87	99.07	49.00
PSO	255,059.75	275,770.43	92.49	8.65	99.54	49.67
GWO	248,256.99	266,827.88	93.04	8.97	99.04	37.00
ACO	78,896.91	80,462.11	98.05	8.54	99.10	99.67
FA	97,303.50	96,633.60	99.31	8.42	99.12	78.00
BA	203,277.77	211,079.46	96.30	8.33	99.12	53.33
ABC	159,359.54	160,944.92	99.01	16.41	99.14	66.67
WOA	85,749.06	73,810.01	83.82	8.19	99.15	99.67

residual variance at higher ETP values. Similar heteroscedastic patterns were observed for Gradient Boosting, XGBoost, LightGBM, and CatBoost, although these ensemble boosting methods substantially reduced error magnitude compared with single-tree and bagging approaches.

Among all evaluated surrogates, the ANN demonstrated superior performance, achieving the lowest residual errors (RMSE = 0.264, MAE = 0.184) and exhibiting homoscedastic residuals across the full ETP range. Importantly, no statistically significant systematic bias was detected in the ANN on the test set (sign test, $p = 0.885$; Wilcoxon, $p = 0.133$), indicating symmetric residuals centered on zero.

These findings were further reinforced in the constrained physiological parameter set (95–105%), where performance differences became more pronounced. The ANN again outperformed all alternatives (RMSE = 0.116,

Table 8 Comparative analysis of surrogate-based optimization versus true ODE simulation using the XGBoost model

MOA	Optimal DMP (Surrogate)	True DMP (Full ODE)	Relative Accuracy (%)	Surrogate average Simulation Time (min)	Time Reduction (%)	Convergence efficiency (%)
GA	132,163.82	120,584.88	90.40	9.97	98.95	75.67
PSO	248,642.43	260,831.49	95.33	10.00	99.46	55.33
GWO	240,565.51	265,926.60	90.46	9.79	98.96	36.67
ACO	79,890.59	71,443.70	88.18	10.20	98.92	99.67
FA	99,276.32	94,890.80	95.38	10.18	98.94	28.00
BA	200,343.25	199,987.94	99.82	10.07	98.93	16.67
ABC	164,303.78	179,937.42	91.31	19.83	98.96	48.00
WOA	87,843.61	78,593.91	88.23	10.13	98.95	99.67

Table 9 Comparative analysis of surrogate-based optimization versus true ODE simulation using the lightGBM model

MOA	Optimal DMP (Surrogate)	True DMP (Full ODE)	Relative Accuracy (%)	Surrogate average Simulation Time (min)	Time Reduction (%)	Convergence efficiency (%)
GA	136,669.38	130,454.01	95.24	7.88	99.17	64.67
PSO	258,027.76	258,823.10	99.69	7.79	99.58	40.33
GWO	260,583.74	278,388.31	93.60	8.38	99.11	31.67
ACO	80,949.25	80,056.92	98.89	7.02	99.26	99.33
FA	100,117.57	101,118.28	99.01	7.10	99.26	65.00
BA	202,503.21	210,327.42	96.28	6.39	99.32	49.00
ABC	176,629.61	187,564.46	94.17	13.69	99.28	18.33
WOA	85,308.19	69,134.20	76.58	6.57	99.32	99.33

Table 10 Comparative analysis of surrogate-based optimization versus true ODE simulation using the CatBoost model

MOA	Optimal DMP (Surrogate)	True DMP (Full ODE)	Relative Accuracy (%)	Surrogate average Simulation Time (min)	Time Reduction (%)	Convergence efficiency (%)
GA	165,160.00	160,875.62	97.41	6.31	99.34	67.33
PSO	243,024.01	250,098.07	97.09	6.04	99.68	39.00
GWO	264,267.71	276,729.70	95.29	6.55	99.30	36.33
ACO	80,567.76	80,462.11	99.87	5.91	99.37	99.67
FA	96,713.00	90,084.13	93.15	6.35	99.34	66.00
BA	209,966.45	207,170.14	98.67	6.52	99.31	33.67
ABC	169,621.07	180,415.22	93.64	12.65	99.33	71.33
WOA	78,512.16	53,243.13	67.84	6.02	99.38	99.33

MAE = 0.094), whereas the second-best model (LightGBM) showed substantially higher errors (RMSE = 5.25, MAE = 4.28). Residual spread plots in this local neighborhood demonstrate the markedly tighter error distribution of the ANN compared with that of tree-based ensembles.

Although boosting-based models such as LightGBM and CatBoost may be considered worse than ANN, their residual biases remained small in absolute terms. For example, mean absolute bias values for LightGBM (≈2.63) and CatBoost (≈2.29) correspond to approximately 0.34% of the mean true ETP value, suggesting limited practical impact.

The residual analyses and non-parametric bias tests consistently indicate that the ANN provides the most accurate and unbiased surrogate across both the global synthetic parameter space and the clinically constrained physiological domain, supporting its suitability for patient-specific coagulation modeling. Boosting models, although

Table 11 Comparative analysis of surrogate-based optimization versus true ODE simulation using the ANN model

MOA	Optimal DMP (Surrogate)	True DMP (Full ODE)	Relative Accuracy (%)	Surrogate average Simulation Time (min)	Time Reduction (%)	Convergence efficiency (%)
GA	138,085.53	116,416.98	84.32	8.35	99.12	72.67
PSO	275,547.26	254,905.28	92.51	8.32	99.55	57.00
GWO	291,759.95	285,828.28	97.97	8.31	99.12	28.33
ACO	82,271.26	71,443.70	86.85	8.10	99.14	99.67
FA	99,395.68	97,140.56	97.73	8.30	99.13	50.67
BA	209,004.03	188,181.20	90.06	8.00	99.15	73.33
ABC	206,077.54	206,147.72	99.97	16.02	99.16	37.33
WOA	79,058.68	53,243.13	67.30	8.24	99.15	99.33

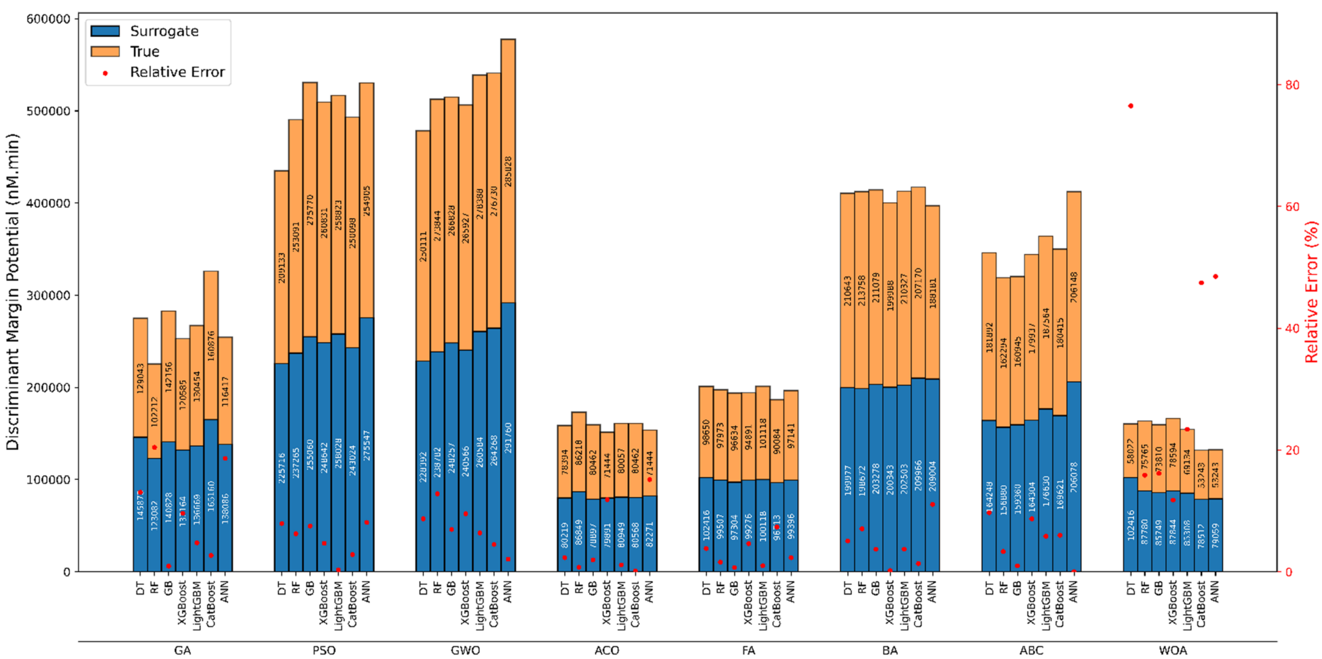

Fig. 6 Performance evaluation of surrogate models across multiple metaheuristic optimization algorithms and machine learning techniques

presenting limitations, remain viable due to their low bias magnitude and were also considered in the following analyses.

Tables 5, 6, 7, 8, 9, 10 and 11 present the comparative performance of surrogate-based optimization using different ML models against the true ODE simulations across multiple MOAs. Across all surrogates, the simulation time was drastically reduced, with time savings exceeding 98% in most cases and reaching over 99% for DT, GB, LightGBM, CatBoost, and ANN models. Relative accuracy varied across models and MOAs, with the highest values (>99%) observed for ACO and FA across several surrogates, while WOA consistently exhibited lower accuracy (as low as 23.46% for the DT model). The RF model, despite maintaining competitive accuracy, incurred substantially higher average simulation times (>300 min) than other surrogates, due to the greater complexity and size of the model (many trees). Among the MOAs analyzed, GWO and PSO achieved the highest DMP values in both surrogate and true ODE predictions, consistently across ML models. The ANN–GWO combination attained the highest objective function value during hybrid model optimization. Although ACO and WOA demonstrated high convergence efficiency, their low DMP values suggest that these MOAs were unable to

Table 12 Binary classification performance of the ANN-Surrogate framework across ML algorithms and optimization methods

ML	MOA	Settings*	AUC Train	ACC Train	TPR Train	TNR Train	F1score Train	AUC Test	ACC Test	TPR Test	TNR Test	F1Score Test
DT	GA	S-U-40-A-S-M	0.80658	0.83030	0.82353	0.83206	0.66667	0.59091	0.68571	0.73333	0.67273	0.50000
DT	PSO	S-U-30-P-S-M	0.92782	0.92121	0.94118	0.91603	0.83117	0.82182	0.87143	0.93333	0.85455	0.75676
DT	GWO	S-U-30-P-C-M	0.85631	0.84242	0.91176	0.82443	0.70455	0.84788	0.87143	0.93333	0.85455	0.75676
DT	ACO	S-U-50-A-S-M	0.53065	0.63030	0.73529	0.60305	0.45045	0.49697	0.60000	0.73333	0.56364	0.44000
DT	FA	S-U-50-A-S-S	0.68321	0.67273	0.70588	0.66412	0.47059	0.71515	0.65714	0.86667	0.60000	0.52000
DT	BA	S-U-30-A-C-F	0.84677	0.81818	0.88235	0.80153	0.66667	0.88303	0.77143	1.00000	0.70909	0.65217
DT	ABC	N-U-50-P-C-F	0.74405	0.83636	0.73529	0.86260	0.64935	0.72485	0.80000	0.66667	0.83636	0.58824
DT	WOA	S-U-50-A-S-S	0.68321	0.67273	0.70588	0.66412	0.47059	0.71515	0.65714	0.86667	0.60000	0.52000
RF	GA	S-U-30-A-S-M	0.74181	0.71515	0.82353	0.68702	0.54369	0.73515	0.68571	0.80000	0.65455	0.52174
RF	PSO	S-U-40-P-C-M	0.90952	0.93333	0.88235	0.94656	0.84507	0.90667	0.88571	0.86667	0.89091	0.76471
RF	GWO	S-U-40-A-C-S	0.92075	0.93939	0.88235	0.95420	0.85714	0.88242	0.84286	0.93333	0.81818	0.71795
RF	ACO	S-U-50-A-C-M	0.68949	0.66667	0.70588	0.65649	0.46602	0.65939	0.68571	0.66667	0.69091	0.47619
RF	FA	S-U-50-P-S-S	0.69275	0.63030	0.82353	0.58015	0.47863	0.76485	0.61429	0.80000	0.56364	0.47059
RF	BA	S-U-20-A-C-S	0.90144	0.86667	0.88235	0.86260	0.73171	0.79879	0.85714	0.80000	0.87273	0.70588
RF	ABC	M-U-50-P-S-F	0.79625	0.82424	0.82353	0.82443	0.65882	0.68727	0.82857	0.66667	0.87273	0.62500
RF	WOA	M-U-50-A-S-M	0.65043	0.67273	0.64706	0.67939	0.44485	0.72848	0.71429	0.80000	0.69091	0.54545
GB	GA	S-U-40-A-C-S	0.79838	0.76364	0.79412	0.75573	0.58065	0.68364	0.62857	0.73333	0.60000	0.45833
GB	PSO	S-U-50-P-C-M	0.95195	0.93333	0.97059	0.92366	0.85714	0.86788	0.85714	0.86667	0.85455	0.72222
GB	GWO	S-U-30-A-C-M	0.93444	0.89697	0.91176	0.89313	0.78481	0.90182	0.85714	0.93333	0.83636	0.73684
GB	ACO	S-U-30-P-C-F	0.66637	0.63636	0.73529	0.61069	0.45455	0.65455	0.61429	0.53333	0.63636	0.30476
GB	FA	S-U-50-A-S-F	0.71778	0.66061	0.82353	0.61832	0.50000	0.63636	0.55714	0.73333	0.50909	0.41509
GB	BA	S-U-20-P-C-F	0.90031	0.86061	0.94118	0.83969	0.73563	0.72000	0.72857	0.80000	0.70909	0.55814
GB	ABC	S-U-50-P-S-F	0.79255	0.71515	0.91176	0.66412	0.56881	0.80121	0.70000	1.00000	0.61818	0.58824
GB	WOA	S-U-40-P-C-F	0.74405	0.64242	0.82353	0.59542	0.48696	0.62424	0.57143	0.66667	0.54545	0.38095
XGBoost	GA	S-U-40-A-S-F	0.72149	0.79394	0.73529	0.80916	0.59524	0.81576	0.74286	0.86667	0.70909	0.59091
XGBoost	PSO	S-U-20-P-C-M	0.93309	0.93939	0.91176	0.94656	0.86111	0.91636	0.88571	0.80000	0.90909	0.75000
XGBoost	GWO	S-U-50-A-C-M	0.93714	0.92727	0.91176	0.93130	0.83784	0.92364	0.88571	0.93333	0.87273	0.77778
XGBoost	ACO	S-U-30-A-S-M	0.70914	0.73333	0.70588	0.74046	0.52174	0.64970	0.64286	0.66667	0.63636	0.44444
XGBoost	FA	S-U-30-P-S-F	0.69533	0.66061	0.76471	0.63359	0.48148	0.63333	0.60000	0.66667	0.58182	0.40404
XGBoost	BA	M-U-30-A-C-M	0.86013	0.84242	0.88235	0.83206	0.69767	0.86061	0.81429	0.93333	0.78182	0.68293
XGBoost	ABC	S-U-30-P-C-F	0.86080	0.83030	0.79412	0.83969	0.65854	0.73697	0.85714	0.53333	0.94545	0.61538
XGBoost	WOA	S-U-50-P-S-S	0.65413	0.69091	0.70588	0.68702	0.48485	0.70970	0.61429	0.66667	0.60000	0.41667
LightGBM	GA	S-U-40-A-C-M	0.80108	0.79394	0.76471	0.80153	0.60465	0.74182	0.65714	0.66667	0.65455	0.45455
LightGBM	PSO	S-U-40-A-C-M	0.95442	0.93333	0.94118	0.93130	0.85333	0.91515	0.87143	0.86667	0.87273	0.74286
LightGBM	GWO	S-U-40-A-C-S	0.91311	0.91515	0.91176	0.91603	0.81579	0.92485	0.87143	0.93333	0.85455	0.75676
LightGBM	ACO	S-U-40-P-S-S	0.68040	0.74545	0.61765	0.77863	0.50000	0.49576	0.58571	0.33333	0.65455	0.13889
LightGBM	FA	S-U-40-A-S-M	0.72654	0.66667	0.79412	0.63359	0.49541	0.60182	0.55714	0.73333	0.50909	0.41509
LightGBM	BA	S-U-40-A-C-F	0.89313	0.87273	0.88235	0.87023	0.74074	0.80606	0.80000	0.93333	0.76364	0.66667
LightGBM	ABC	N-U-40-P-C-F	0.83330	0.84848	0.76471	0.87023	0.67532	0.78000	0.80000	0.80000	0.80000	0.63158
LightGBM	WOA	S-U-30-P-S-F	0.69241	0.69091	0.76471	0.67176	0.50485	0.60000	0.55714	0.46667	0.58182	0.21778

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Table 12 (continued)

ML	MOA	Settings*	AUC Train	ACC Train	TPR Train	TNR Train	F1score Train	AUC Test	ACC Test	TPR Test	TNR Test	F1Score Test
CatBoost	GA	S-U-40-A-C-S	0.82196	0.86667	0.70588	0.90840	0.68571	0.73091	0.81429	0.60000	0.87273	0.58065
CatBoost	PSO	S-U-30-P-C-M	0.91154	0.85455	0.97059	0.82443	0.73333	0.90061	0.84286	0.93333	0.81818	0.71795
CatBoost	GWO	S-U-40-P-C-M	0.92187	0.89091	0.97059	0.87023	0.78571	0.97091	0.91429	1.00000	0.89091	0.83333
CatBoost	ACO	S-U-30-P-C-F	0.68006	0.63636	0.73529	0.61069	0.45455	0.67030	0.65714	0.66667	0.65455	0.45455
CatBoost	FA	S-U-40-P-S-M	0.71082	0.67879	0.73529	0.66412	0.48544	0.62970	0.57143	0.73333	0.52727	0.42308
CatBoost	BA	S-U-20-A-C-F	0.82914	0.88485	0.76471	0.91603	0.73239	0.88485	0.87143	0.80000	0.89091	0.72727
CatBoost	ABC	S-U-20-P-C-F	0.87057	0.76364	0.94118	0.71756	0.62136	0.81212	0.71429	0.86667	0.67273	0.56522
CatBoost	WOA	N-U-20-A-S-F	0.65963	0.73333	0.61765	0.76336	0.48837	0.72364	0.68571	0.66667	0.69091	0.47619
ANN	GA	S-U-40-P-S-M	0.78682	0.72121	0.94118	0.66412	0.58182	0.76727	0.62857	0.80000	0.58182	0.48000
ANN	PSO	S-U-20-P-C-M	0.87315	0.92121	0.88235	0.93130	0.82192	0.93939	0.88571	0.93333	0.87273	0.77778
ANN	GWO	S-U-40-P-C-M	0.96677	0.94545	0.94118	0.94656	0.87671	0.91636	0.90000	0.93333	0.89091	0.80000
ANN	ACO	S-U-30-A-S-M	0.69903	0.70303	0.73529	0.69466	0.50505	0.67152	0.61429	0.66667	0.60000	0.41667
ANN	FA	S-U-20-P-C-F	0.67041	0.71515	0.70588	0.71756	0.50526	0.85212	0.74286	0.73333	0.74545	0.55000
ANN	BA	S-U-20-A-C-F	0.89672	0.87879	0.82353	0.89313	0.73684	0.84970	0.82857	0.86667	0.81818	0.68421
ANN	ABC	N-U-40-P-C-F	0.84564	0.78788	0.88235	0.76336	0.63158	0.82303	0.71429	0.86667	0.67273	0.56522
ANN	WOA	N-U-20-A-S-F	0.65065	0.73333	0.61765	0.76336	0.48837	0.69576	0.68571	0.66667	0.69091	0.47619

*Model specifications (see Table 2) are abbreviated in the format α - β - γ - δ - ε – ζ, where: α represents the patient input data normalization scheme ("S" = standard; "M" = MinMax). β represents the initial state specification ("P" = personalized; "U" = uniform). γ represents the neural network width (20, 30, 40, or 50 neurons in the hidden layer). δ represents the dimensionality reduction strategy ("A" = all features; "P" = PCA-based variables). ε represents the model target scope ("S" = selective, use a subset identified as most influential through sensitivity analysis; "C" = comprehensive, use the complete set of kinetic parameters). ζ represents optimization criterion ("M" = margin-only, use only DMP; "S" = Sparse, combines DMP with L1 regularization; "F" = Full regularization, combines DMP with both L1 and L2 penalties) improve the model, indicating limited optimization capability under these conditions. In contrast, PSO and GWO exhibited moderate convergence efficiency but achieved higher DMP values, indicating a more effective balance between exploitation and exploration.

Figure 6 graphically illustrates the performance differences among the ML–MOA combinations. PSO and GWO outperformed all other optimizers across the seven top ML models. BA and ABC achieved intermediate results, while GA, ACO, FA, and WOA showed comparatively lower performance. Among all configurations, ANN–GWO achieved the highest DMP value, closely followed by ANN–PSO, reinforcing the strong compatibility between these optimizers and the ANN architecture.

The comparative analysis of Tables 12 and 13 highlights evident performance disparities among the evaluated MOAs. Across both the surrogate and true ANN–ODE frameworks, ACO and WOA frequently ranked among the lowest-performing optimizers, with lower AUC, ACC, and F1-score values across most ML algorithms, indicating limited capacity to effectively fine-tune model parameters. In contrast, GWO and PSO demonstrated remarkable efficiency on the test set, achieving superior predictive accuracy and robustness, particularly when coupled with boosting-based ML models (GB, XGBoost, LightGBM, and CatBoost) and ANN. These combinations not only yielded the highest AUC values, frequently exceeding 0.90, but also maintained balanced sensitivity and specificity, reflecting strong generalization and stability. Notably, the ANN models optimized by GWO achieved peak test performance, indicating strong compatibility between GWO optimization and the ANN architecture.

Regarding model configurations, Most top-ranked configurations employed standardized (Z-score) inputs, with most top-ranked models using Zhu's [35] baseline blood concentrations. The network widths varied across ML–MOA combinations, indicating no consistent preference for a specific hidden-layer size. Notably, the ANN-based hybrids with GWO and PSO achieved superior predictive performance when paired with PCA-based dimensionality reduction and patient-specific kinetic parameter calibration, with optimization restricted to the DMP-only objective. Configurations optimized using the DMP-only objective frequently demonstrated higher test performance than sparsity- or fully regularized schemes. Across other ML families (DT, RF, GB, XGBoost, LightGBM, and CatBoost), no single configuration dominated. However, models employing PCA-reduced features and comprehensive adjustments to kinetic parameters tended to outperform those trained on all raw variables. In particular, PSO- and GWO-driven surrogates repeatedly ranked among the highest in AUC and ACC, suggesting that swarm-based search strategies were more effective in navigating adjustable parameter space. Conversely, ACO, FA, and WOA configurations showed weaker stability across both ML and ANN surrogates, with performance sensitive to feature selection and regularization strategy. Overall, these results highlight the importance of both dimensionality reduction and objective function design in tailoring surrogate model efficiency, with ANN-PSO and ANN-GWO emerging as the most robust patient-specific predictors.

To evaluate the discriminatory performance of different MOA-ML models in separating patients with RVTE from those without RVTE, we computed the mean ETP for each clinical group. For each model, the separation between RVTE and non-RVTE patients was defined as the difference between the mean ETP of the two groups. The reference model, ANN-GWO, was compared to each alternative MOA-ML model by calculating the difference in separation (ANN-GWO minus the comparative model). Statistical significance and uncertainty were assessed using a bootstrap-based one-sided Wilcoxon signed-rank approach. Specifically, RVTE and non-RVTE ETP values from the reference model were resampled with replacement 10,000 times to generate a distribution of bootstrap separation estimates. For each bootstrap sample, the difference between the reference separation and the comparative model's separation was calculated, yielding a bootstrap distribution of differences. The one-sided p-value was computed as the proportion of bootstrap differences less than or equal to zero, reflecting the probability that the reference model is not superior. The 95% CI for the median difference was estimated as the 2.5th and 97.5th percentiles of the bootstrap distribution. Positive median differences indicate superior discrimination by ANN-GWO, and statistical significance was defined as $p < 0.05$. This nonparametric bootstrap approach allows robust inference without assuming normality and provides a reliable assessment of differences in model performance.

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Table 13 Binary classification performance of the true ANN–ODE framework using the adjustable parameters optimized via the ANN-Surrogate framework across ML models and MOAs

ML	MOA	AUC Train	ACC Train	TPR Train	TNR Train	F1score Train	AUC Test	ACC Test	TPR Test	TNR Test	F1Score Test
DT	GA	0.79277	0.83030	0.82353	0.83206	0.66667	0.65091	0.68571	0.73333	0.67273	0.50000
DT	PSO	0.91693	0.92121	0.94118	0.91603	0.83117	0.86788	0.87143	0.93333	0.85455	0.75676
DT	GWO	0.82645	0.84242	0.91176	0.82443	0.70455	0.88848	0.87143	0.93333	0.85455	0.75676
DT	ACO	0.65739	0.59394	0.82353	0.53435	0.45528	0.62909	0.51429	0.80000	0.43636	0.41379
DT	FA	0.68321	0.67273	0.70588	0.66412	0.47059	0.69576	0.65714	0.86667	0.60000	0.52000
DT	BA	0.86439	0.85455	0.82353	0.86260	0.70000	0.87394	0.78571	0.93333	0.74545	0.65116
DT	ABC	0.77930	0.83636	0.73529	0.86260	0.64935	0.78667	0.80000	0.66667	0.83636	0.58824
DT	WOA	0.70117	0.73333	0.67647	0.74809	0.51111	0.45576	0.57143	0.40000	0.61818	0.17778
RF	GA	0.73664	0.71515	0.82353	0.68702	0.54369	0.62182	0.68571	0.80000	0.65455	0.52174
RF	PSO	0.89717	0.92121	0.88235	0.93130	0.82192	0.88727	0.87143	0.86667	0.87273	0.74286
RF	GWO	0.93534	0.93939	0.88235	0.95420	0.85714	0.85939	0.82857	0.86667	0.81818	0.68421
RF	ACO	0.69511	0.66061	0.70588	0.64885	0.46154	0.64121	0.67143	0.60000	0.69091	0.41538
RF	FA	0.71599	0.63030	0.82353	0.58015	0.47863	0.69333	0.61429	0.80000	0.56364	0.47059
RF	BA	0.89021	0.87273	0.88235	0.87023	0.74074	0.79152	0.84286	0.73333	0.87273	0.66667
RF	ABC	0.76134	0.81818	0.82353	0.81679	0.65116	0.72364	0.80000	0.66667	0.83636	0.58824
RF	WOA	0.63247	0.67273	0.64706	0.67939	0.44485	0.76121	0.71429	0.80000	0.69091	0.54545
GB	GA	0.79300	0.85455	0.61765	0.91603	0.63636	0.70667	0.71429	0.53333	0.76364	0.40635
GB	PSO	0.94612	0.91515	1.00000	0.89313	0.82927	0.85212	0.85714	0.86667	0.85455	0.72222
GB	GWO	0.90840	0.87879	0.94118	0.86260	0.76190	0.90303	0.81429	0.93333	0.78182	0.68293
GB	ACO	0.67894	0.66061	0.70588	0.64885	0.46154	0.66424	0.62857	0.53333	0.65455	0.31605
GB	FA	0.70566	0.63030	0.82353	0.58015	0.47863	0.67030	0.58571	0.80000	0.52727	0.45283
GB	BA	0.87741	0.84848	0.94118	0.82443	0.71910	0.71758	0.72857	0.80000	0.70909	0.55814
GB	ABC	0.80624	0.72727	0.91176	0.67939	0.57944	0.78909	0.70000	0.80000	0.67273	0.53333
GB	WOA	0.74225	0.64242	0.82353	0.59542	0.48696	0.57333	0.57143	0.66667	0.54545	0.38095
XGBoost	GA	0.72474	0.79394	0.73529	0.80916	0.59524	0.72000	0.74286	0.86667	0.70909	0.59091
XGBoost	PSO	0.91491	0.93333	0.91176	0.93893	0.84932	0.87758	0.87143	0.73333	0.90909	0.70968
XGBoost	GWO	0.91109	0.91515	0.91176	0.91603	0.81579	0.87515	0.85714	0.93333	0.83636	0.73684
XGBoost	ACO	0.74091	0.69697	0.73529	0.68702	0.50000	0.54545	0.58571	0.66667	0.56364	0.39216
XGBoost	FA	0.64212	0.65455	0.79412	0.61832	0.48649	0.58182	0.54286	0.66667	0.50909	0.36036
XGBoost	BA	0.86866	0.82424	0.91176	0.80153	0.68132	0.84242	0.75714	0.93333	0.70909	0.62222
XGBoost	ABC	0.86080	0.78182	0.85294	0.76336	0.61702	0.74182	0.74286	0.73333	0.74545	0.55000
XGBoost	WOA	0.65447	0.69091	0.70588	0.68702	0.48485	0.68727	0.61429	0.66667	0.60000	0.41667
LightGBM	GA	0.80736	0.84848	0.76471	0.87023	0.67532	0.71515	0.72857	0.60000	0.76364	0.48649
LightGBM	PSO	0.93983	0.92121	0.97059	0.90840	0.83544	0.88485	0.85714	0.86667	0.85455	0.72222
LightGBM	GWO	0.90391	0.91515	0.91176	0.91603	0.81579	0.90788	0.87143	0.93333	0.85455	0.75676
LightGBM	ACO	0.66300	0.66061	0.70588	0.64885	0.46154	0.52485	0.57143	0.66667	0.54545	0.38095
LightGBM	FA	0.71711	0.69697	0.76471	0.67939	0.50980	0.63030	0.61429	0.66667	0.60000	0.41667
LightGBM	BA	0.88258	0.81212	0.91176	0.78626	0.66667	0.81091	0.75714	0.93333	0.70909	0.62222
LightGBM	ABC	0.82869	0.83636	0.76471	0.85496	0.65823	0.79152	0.78571	0.73333	0.80000	0.59459
LightGBM	WOA	0.67423	0.65455	0.82353	0.61069	0.49558	0.62545	0.55714	0.66667	0.52727	0.37037
CatBoost	GA	0.82645	0.88485	0.67647	0.93893	0.70769	0.73333	0.77143	0.46667	0.85455	0.43556
CatBoost	PSO	0.87427	0.86061	0.94118	0.83969	0.73563	0.85697	0.82857	0.86667	0.81818	0.68421
CatBoost	GWO	0.92995	0.88485	0.97059	0.86260	0.77647	0.91879	0.88571	1.00000	0.85455	0.78947
CatBoost	ACO	0.67894	0.66061	0.70588	0.64885	0.46154	0.66424	0.62857	0.53333	0.65455	0.31605
CatBoost	FA	0.69578	0.67879	0.73529	0.66412	0.48544	0.56121	0.57143	0.73333	0.52727	0.42308
CatBoost	BA	0.82600	0.83030	0.79412	0.83969	0.65854	0.87394	0.82857	0.80000	0.83636	0.66667
CatBoost	ABC	0.88348	0.75758	0.94118	0.70992	0.61538	0.81091	0.68571	0.86667	0.63636	0.54167
CatBoost	WOA	0.63583	0.73333	0.61765	0.76336	0.48837	0.64727	0.68571	0.66667	0.69091	0.47619
ANN	GA	0.76134	0.72121	0.94118	0.66412	0.58182	0.67879	0.62857	0.80000	0.58182	0.48000
ANN	PSO	0.84890	0.91515	0.88235	0.92366	0.81081	0.92364	0.82857	0.86667	0.81818	0.68421
ANN	GWO	0.93534	0.94545	0.94118	0.94656	0.87671	0.89455	0.90000	0.93333	0.89091	0.80000
ANN	ACO	0.74091	0.69697	0.73529	0.68702	0.50000	0.54545	0.58571	0.66667	0.56364	0.39216

Table 13 (continued)

ML	MOA	AUC Train	ACC Train	TPR Train	TNR Train	F1score Train	AUC Test	ACC Test	TPR Test	TNR Test	F1Score Test
ANN	FA	0.68635	0.70909	0.70588	0.70992	0.50000	0.83030	0.75714	0.80000	0.74545	0.58537
ANN	BA	0.88617	0.84242	0.85294	0.83969	0.69048	0.84848	0.72857	0.93333	0.67273	0.59574
ANN	ABC	0.83767	0.78182	0.88235	0.75573	0.62500	0.81697	0.71429	0.86667	0.67273	0.56522
ANN	WOA	0.63583	0.73333	0.61765	0.76336	0.48837	0.64727	0.68571	0.66667	0.69091	0.47619

As shown in Table 14, ANN-GWO consistently achieved greater mean ETP separation than most competing ML–MOA combinations, with statistically significant superiority ($p < 0.05$) in the majority of pairwise comparisons, particularly against GA-, ACO-, FA-, BA-, ABC-, and WOA-based models. In statistically significant comparisons, confidence intervals were largely positive, supporting the stability of the observed differences. Overall, ANN-GWO demonstrated strong discrimination between RVTE and non-RVTE patients. While ANN-GWO attained the highest DMP among surrogate models, its advantage over other ML surrogates with GWO was not statistically significant. Nonetheless, pairing GWO with boosting-based surrogates yielded competitive results, except for RF. RF-based surrogates, although competitive in discrimination, were previously shown to offer only a limited computational advantage over ANN and boosting models. The similar performance trends observed for GWO and PSO suggest that both MOAs are highly effective optimizers within this hybrid framework.

Figure 7 shows the ROC curves for the test set, where predictions were generated using the ANN in place of the ODE component of the hybrid framework. Among the optimizers, PSO (AUC = 0.939) and GWO (AUC = 0.916) exhibited the highest discriminatory power, markedly outperforming the others. In contrast, ACO (AUC = 0.672) and WOA (AUC = 0.696) demonstrated substantially lower performance. Figure 8 presents the ROC curves obtained with the optimally tuned parameters in the hybrid framework with ODE, showing similar trends but slightly lower AUC values. These findings indicate that PSO and GWO consistently maintain superior performance, even within the computationally efficient surrogate framework, demonstrating stable performance across surrogate and ODE-based implementations.

Figure 9 presents radar plots summarizing the performance metrics of different MOAs across the seven best ML models (DT, RF, GB, XGBoost, LightGBM, CatBoost, and ANN). Across most models, PSO and GWO consistently exhibit superior performance, achieving high relative accuracy, AUC, TPR, TNR, and F1-scores, while also maintaining strong convergence efficiency and time reduction. GA and BA perform competitively in certain ML frameworks, yet they generally lag behind PSO and GWO. In contrast, ABC and WOA demonstrate lower normalized true performance and relative accuracy across multiple metrics, particularly in models such as DT and GB. Optimizers like ACO achieve moderate accuracy and discriminatory power, but with comparatively lower convergence efficiency. These trends highlight that PSO and GWO provide a balanced combination of predictive performance and computational efficiency across ML frameworks, whereas other MOAs often trade off accuracy for speed or vice versa. The impact of ML integration is most evident in computational time, with notable reductions across most methods. However, RF exhibited comparatively smaller time savings. In terms of predictive relative accuracy, binary classification metrics remained close to 100% for PSO- and GWO-based models, and were consistently high for BA and FA. By contrast, other MOAs showed greater variability and larger errors across ML algorithms, including WOA with DT, GA with GB, ABC with XGBoost, and ACO with LightGBM.

Figure 10 presents radar plots of normalized performance metrics across different ML models under various MOAs. Overall, the plots reveal distinct performance patterns for each MOA. GWO- and PSO-based models consistently exhibit high normalized true performance and relative accuracy across all metrics, particularly excelling in relative aaccuracies of AUC, ACC, and F1-score for ANN and boosting models. Conversely, ABC- and WOA-based models generally demonstrate lower overall performance, with WOA showing notable reductions in normalized relative accuracy for TPR and F1-score. Intermediate behavior is observed for GA, ACO, FA, and BA, with some metrics approaching peak values while others lag. ANN and GB models maintain robust performance across most MOAs, highlighting their effectiveness when coupled with high-performing optimizers.

Table 14 Comparative performance of ANN-GWO and alternative ML-MOA models in discriminating RVTE from non-RVTE patients using endogenous thrombin potential

MOA	ML model	Median difference (ETP)[1]	*p*-value[2]	95% CI Lower[3]	95% CI Upper[3]
GA	DT	961.53	< 0.0001	621.45	1193.7
GA	RF	1034.7	< 0.0001	702.64	1264.5
GA	GB	918.41	< 0.0001	590.12	1150.0
GA	XGBoost	820.54	< 0.0001	478.24	1054.4
GA	LightGBM	1041.5	< 0.0001	713.27	1278.1
GA	CatBoost	832.85	< 0.0001	498.62	1068.3
GA	ANN	975.56	< 0.0001	644.63	1212.8
PSO	DT	403.72	0.0112	63.506	640.65
PSO	RF	201.17	**0.1066**	−130.55	433.40
PSO	GB	147.95	**0.1758**	−191.74	374.54
PSO	XGBoost	161.61	**0.1613**	−180.1	396.48
PSO	LightGBM	202.79	**0.1059**	−130.29	432.63
PSO	CatBoost	172.16	**0.1428**	−162.15	399.79
PSO	ANN	34.347	**0.4082**	−300.7	265.92
GWO	DT	-0.72527	**0.5023**	-337.8	234.20
GWO	RF	96.585	**0.2691**	-239.16	328.06
GWO	GB	79.407	**0.2995**	-253.18	311.22
GWO	XGBoost	96.002	**0.2729**	-240.36	326.09
GWO	LightGBM	19.73	**0.4460**	-312.92	247.86
GWO	CatBoost	-66.102	**0.6851**	-399.07	168.39
ACO	DT	1024.5	< 0.0001	684.73	1254.0
ACO	RF	999.55	< 0.0001	664.1	1233.6
ACO	GB	1042.8	< 0.0001	705.91	1275.1
ACO	XGBoost	1315	< 0.0001	986.4	1545.1
ACO	LightGBM	1128.4	< 0.0001	787.06	1359.6
ACO	CatBoost	1043.5	< 0.0001	718.56	1275.6
ACO	ANN	1315.9	< 0.0001	972.88	1545.3
FA	DT	844.66	0.0001	512.87	1081.3
FA	RF	934.97	< 0.0001	593.66	1168.2
FA	GB	996.42	< 0.0001	656.59	1227.4
FA	XGBoost	1023.2	< 0.0001	681.59	1255.1
FA	LightGBM	1066.8	< 0.0001	732.25	1302.1
FA	CatBoost	1064.2	< 0.0001	726.04	1295.2
FA	ANN	728.67	0.0001	395.53	967.99
BA	DT	313.92	0.0324	-18.459	545.12
BA	RF	504.11	0.0025	169.46	738.09
BA	GB	583.82	0.0009	242.58	818.87
BA	XGBoost	370.33	0.0196	24.148	604.98
BA	LightGBM	404.07	0.0099	67.231	637.63
BA	CatBoost	243.91	**0.0721**	-95.938	475.04
BA	ANN	487.85	0.0040	148.01	723.09
ABC	DT	449.28	0.0048	114.37	682.88
ABC	RF	655.00	0.0004	322.24	884.80
ABC	GB	562.66	0.0009	231.6	789.81
ABC	XGBoost	621.29	0.0003	283.42	856.89
ABC	LightGBM	570.12	0.0012	237.87	802.33
ABC	CatBoost	594.32	0.0008	260.09	828.41
ABC	ANN	298.87	0.0354	-30.815	530.01
WOA	DT	1455.6	< 0.0001	1121.4	1688.2
WOA	RF	855.41	< 0.0001	512.43	1093.7
WOA	GB	1250.2	< 0.0001	918.44	1483.5
WOA	XGBoost	1067.5	< 0.0001	736.39	1299.2
WOA	LightGBM	1190.2	< 0.0001	853.88	1419.8

Table 14 (continued)

MOA	ML model	Median difference (ETP)[1]	p-value[2]	95% CI Lower[3]	95% CI Upper[3]
WOA	CatBoost	1145.4	< 0.0001	811.34	1378.2
WOA	ANN	1148	< 0.0001	811.44	1382.1

[1]Median difference: Difference in mean ETP separation between RVTE and non-RVTE patients (ANN-GWO minus comparative model)

[2]p-value: One-sided Wilcoxon signed-rank test assessing whether ANN-GWO separation is significantly greater than the comparative model. P-values < 0.05 denote statistically significant superiority of ANN-GWO

[3]95% CI: Percentile-based bootstrap confidence interval of the median difference (10,000 resamples). Positive values indicate that ANN-GWO provides superior discrimination

Bold values indicate non-signifi cant p-values

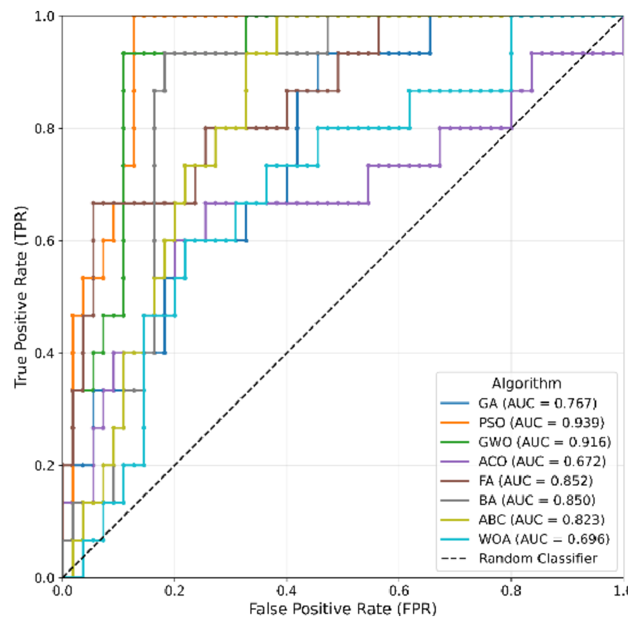

Fig. 7 ROC curves for the test set of the hybrid model with ANN–Surrogate replacing the ODE system, optimized by different MOAs

These results underscore the influence of MOA selection on ML model performance and demonstrate that GWO and PSO are particularly efficient for both classical and ensemble ML approaches.

Together with previous analyses, these visual summaries further support the consistent superiority of swarm-based optimizers (PSO and GWO) in achieving stable, high-accuracy, and computationally efficient hybrid models, particularly when coupled with the ANN surrogate.

3.2 Detailed evaluation of the best-performing ANN surrogate

Based on the previously presented comparative results, the ANN emerged as the most accurate surrogate model. This subsection provides a deeper evaluation of ANN-based models optimized using all MOAs, with particular emphasis on GWO and PSO, which demonstrated the strongest optimization performance.

Following the replacement of the ODE-based kinetic model with a fully differentiable ANN surrogate, gradient-based optimizers became applicable in theory. Accordingly, Adam, RMSProp, and SGD were implemented as baseline optimizers and directly compared with PSO and GWO in terms of convergence behavior, computational efficiency, and predictive fidelity.

Figure 11 illustrates convergence trajectories under both surrogate-based and true ODE evaluations. Because gradient-based methods were approximately three times faster per iteration than MOAs, their iteration budget was increased to 900 (versus 300 for MOAs) to ensure comparable total runtime. Although SGD converged rapidly, it reached substantially lower DMP values, indicating convergence to a local optimum. Adam and RMSProp

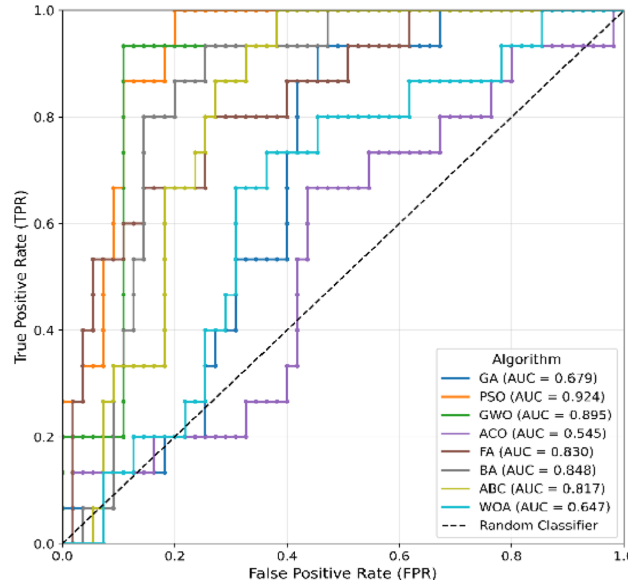

Fig. 8 ROC curves for the test of the true hybrid ANN–ODE model using adjustable parameters optimized via ANN–Surrogate across different MOAs

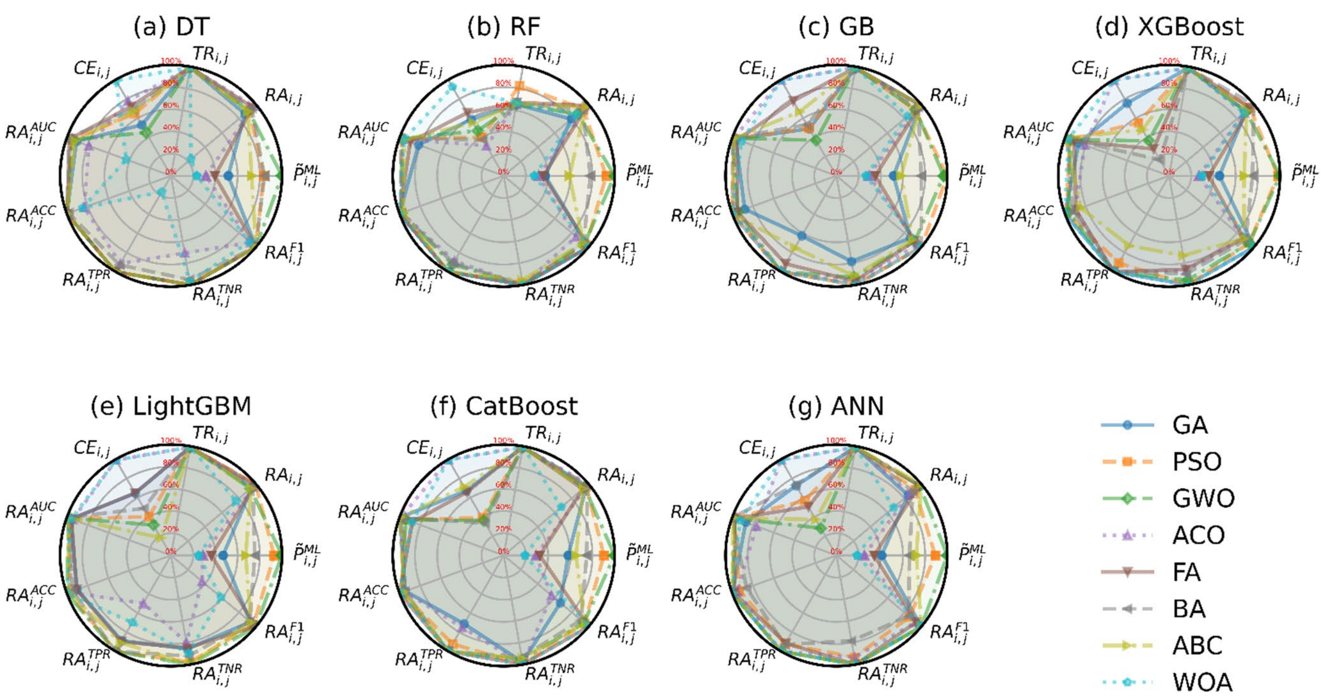

Fig. 9 Radar plots of performance metrics (see Table 3) comparing MOAs across ML models. Metrics include: $\overset{\sim ML}{P_{i,j}}$, $RA_{i,j}$, $TR_{i,j}, CE_{i,j}, RA_{i,j}^{AUC}, RA_{i,j}^{ACC}, RA_{i,j}^{TPR}, RA_{i,j}^{TNR}, RA_{i,j}^{F1}$ for MOAs $j \in \{GA, PSO, GWO, ACO, FA, BA, ABC, WOA\}$ under ML models. (**a**) DT, (**b**) RF, (**c**) GB, (**d**) XGBoost, (**e**) LightGBM, (**f**) CatBoost, and (**g**) ANN

exhibited smoother convergence behavior but showed marked discrepancies between surrogate and true ODE evaluations. In contrast, PSO and GWO consistently achieved higher DMP values and demonstrated strong agreement between surrogate and true evaluations.

Table 15 summarizes the quantitative comparison. While gradient-based optimizers achieved substantial reductions in optimization time (≈99.8%), this improvement was accompanied by lower objective quality. Adam and RMSProp yielded relative accuracies of approximately 66–67%, whereas PSO and GWO maintained accuracies

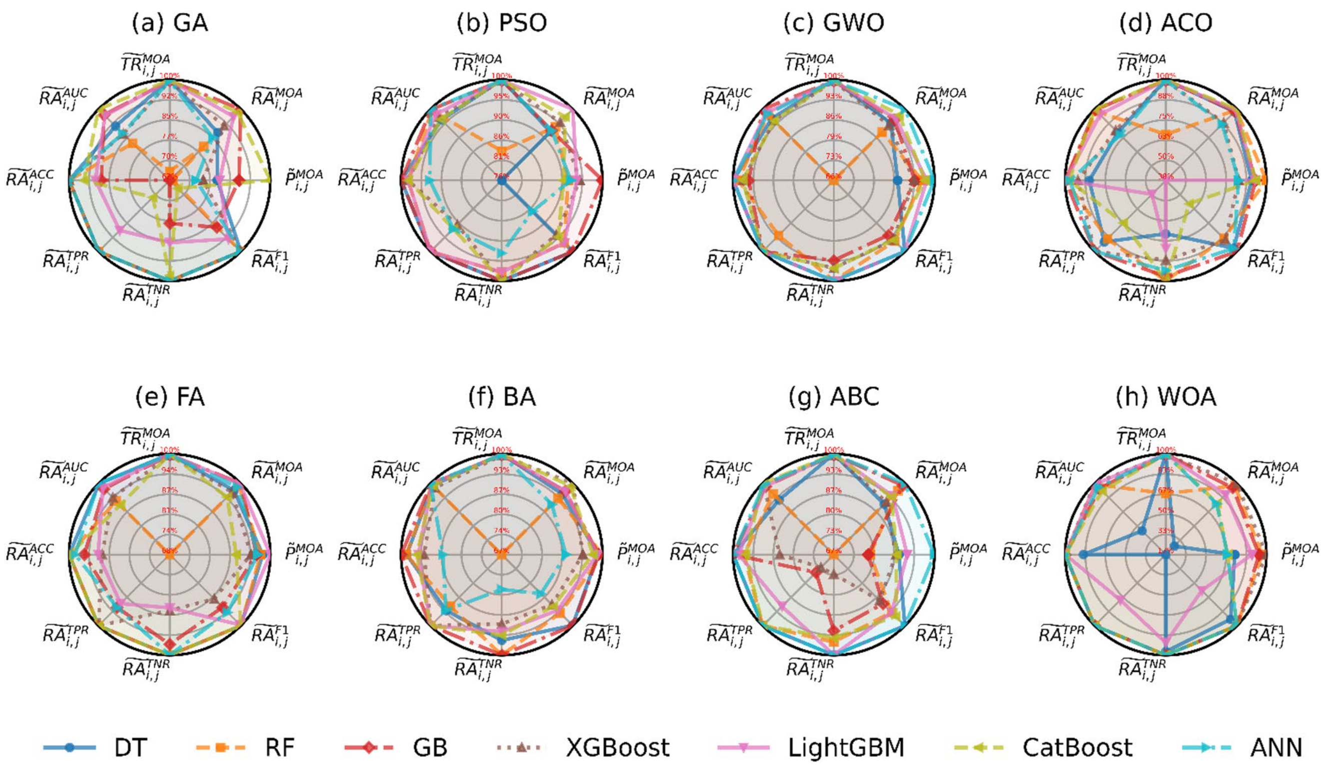

Fig. 10 Radar plots of performance metrics (see Table 3) comparing ML models across MOAs. Metrics include: $\tilde{P}^{MOA}_{i,j}$, $\tilde{RA}^{MOA}_{i,j}$, $\tilde{TR}^{MOA}_{i,j}$, $\tilde{RA}^{AUC}_{i,j}$, $\tilde{RA}^{ACC}_{i,j}$, $\tilde{RA}^{TPR}_{i,j}$, $\tilde{RA}^{TNR}_{i,j}$, $\tilde{RA}^{F1}_{i,j}$ for ML models $i \in \{DT, RF, GB, XGBoost, LightGBM, CatBoost, ANN\}$ under MOAs. (**a**) GA, (**b**) PSO, (**c**) GWO, (**d**) ACO, (**e**) FA, (**f**) BA, (**g**) ABC, and (**h**) WOA

above 92%, with GWO approaching 98%. Although SGD achieved high relative accuracy (99.83%), this reflects strong surrogate–ODE agreement at a low-quality solution, rather than identification of a globally competitive optimum.

Collectively, these findings suggest that, despite differentiability, the ANN-based objective landscape remains highly nonconvex and challenging to optimize. Gradient-based methods appear sensitive to initialization and may converge prematurely, whereas metaheuristic algorithms provide more robust global search and consistently identify higher-quality optima. Therefore, metaheuristic optimization remains advantageous within this hybrid modeling framework.

Table 16 summarizes the predictive performance of ANN surrogate models optimized using different metaheuristic and gradient-based algorithms, while Table 17 reports the corresponding results obtained using the original ODE-based model. Overall, ANN surrogates consistently matched or outperformed the ODE-based counterparts across most metrics and optimization strategies, particularly on the independent test set.

Among all optimization algorithms, the GWO achieved the strongest and most stable performance for the ANN surrogate, yielding the highest test AUC (0.916, 95% CI: 0.840–0.974), along with balanced sensitivity (TPR = 0.933) and specificity (TNR = 0.891). PSO ranked second, demonstrating excellent generalization with a test AUC of 0.939 (95% CI: 0.876–0.985) and high accuracy (0.886), confirming its robustness as an alternative optimizer.

In contrast, gradient-based optimizers (Adam, RMSProp, SGD) showed competitive training performance but substantial degradation on the test set, particularly in sensitivity and F1-score, indicating overfitting and reduced

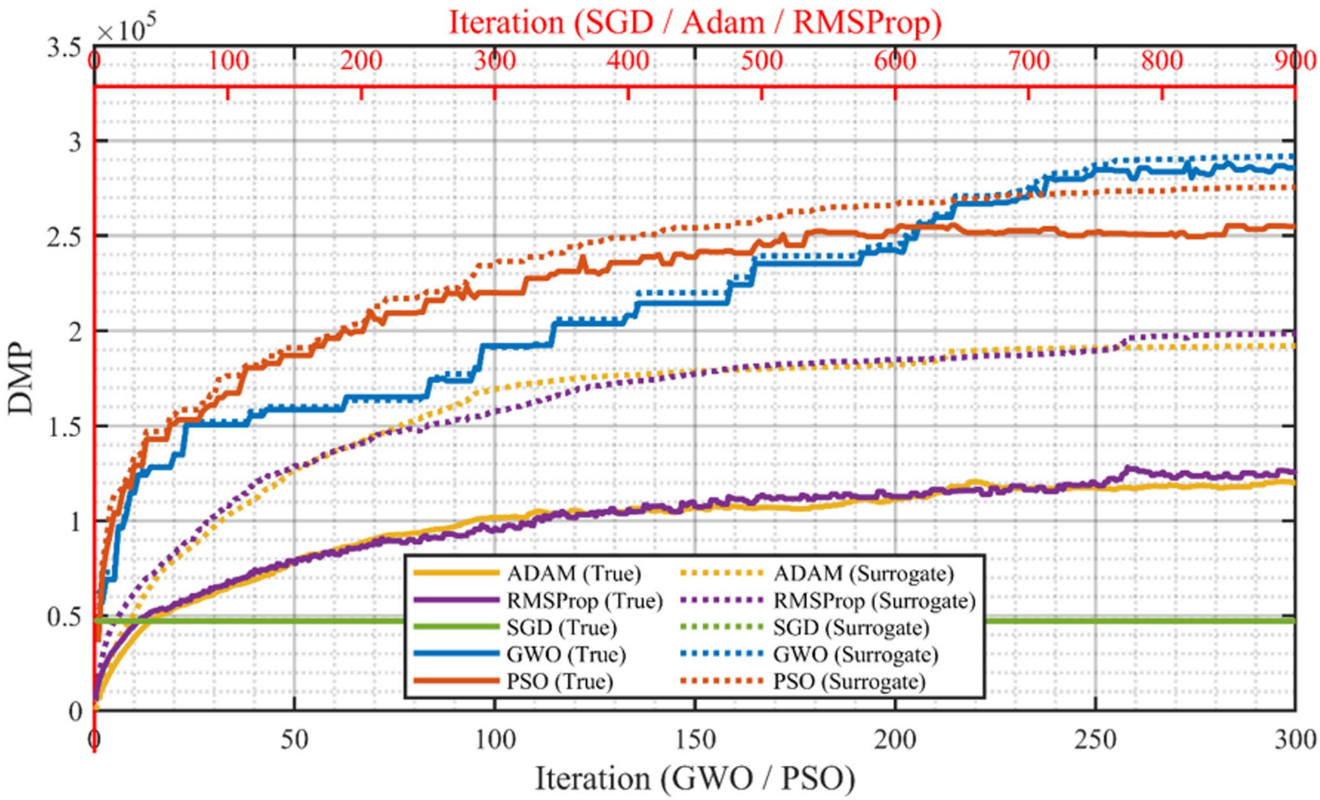

Fig. 11 Convergence behavior of gradient-based and metaheuristic optimizers using ANN surrogate and true ODE evaluations

Table 15 Performance comparison of gradient-based and metaheuristic optimization strategies

MOA	Optimal DMP (Surrogate)	True DMP (Full ODE)	Relative Accuracy (%)	Surrogate average Simulation Time (min)	Time Reduction (%)	Convergence efficiency (%)
PSO	275,547.26	254,905.28	92.51	8.32	99.55	57.00
GWO	291,759.95	285,828.28	97.97	8.31	99.12	28.33
Adam	213,409.55	141,536.55	66.32	2.80	99.80	65.33
RMSProp	219,946.57	148,127.41	67.35	2.74	99.89	50.44
SGD	68,722.41	68,606.05	99.83	2.75	99.88	99.78

generalization. Across nearly all optimization algorithms, ANN-based surrogates achieved test AUCs equal to or higher than those of their ODE-based counterparts. These findings identify the ANN–PSO and ANN–GWO configurations as the most reliable and generalizable modeling strategies, with minimal performance loss upon replacing the ODE model with the ANN surrogate.

Framework sensitivity analysis identified a small subset of design choices with statistically significant and practically meaningful effects on RVTE prediction performance (Table 18). Across metaheuristic optimizers, the normalization strategy consistently emerged as the dominant determinant of test-set AUC, explaining up to 28.9% of the explainable variance and yielding absolute performance gains of up to 0.075. This effect was particularly pronounced for GWO and PSO, which also achieved the highest mean test AUCs under standardized normalization. The feature selection strategy showed a smaller but reproducible influence, with full or PCA-based representations providing modest improvements, depending on the optimizer. Model capacity, reflected by the number of hidden neurons, contributed a secondary yet significant effect in selected algorithms. In contrast, neither the initial condition model nor the kinetic parameter configuration exerted a consistent or substantial

Table 16 Performance of ANN surrogate models optimized by metaheuristic and gradient-based algorithms (bootstrap median and 95% CIs for training and testing metrics)

MOA	AUC Train	ACC Train	TPR Train	TNR Train	F1score Train	AUC Test	ACC Test	TPR Test	TNR Test	F1Score Test
GA	0.7868 (0.7022, 0.8614)	0.7215 (0.6545, 0.7879)	0.9415 (0.8485, 1.0000)	0.6642 (0.5809, 0.7441)	0.5806 (0.4660, 0.6863)	0.7681 (0.6341, 0.8824)	0.6284 (0.5143, 0.7429)	0.8008 (0.5714, 1.0000)	0.5816 (0.4483, 0.7091)	0.4684 (0.2405, 0.6415)
PSO	0.8732 (0.7695, 0.9595)	0.9210 (0.8788, 0.9576)	0.8824 (0.7619, 0.9737)	0.9310 (0.8842, 0.9704)	0.8195 (0.7143, 0.9091)	0.9394 (0.8763, 0.9845)	0.8861 (0.8000, 0.9571)	0.9341 (0.7857, 1.0000)	0.8731 (0.7778, 0.9592)	0.7736 (0.6000, 0.9130)
GWO	0.9673 (0.9172, 0.9961)	0.9456 (0.9091, 0.9758)	0.9419 (0.8495, 1.0000)	0.9466 (0.9051, 0.9841)	0.8753 (0.7838, 0.9493)	0.9164 (0.8400, 0.9740)	0.9001 (0.8286, 0.9571)	0.9327 (0.7778, 1.0000)	0.8912 (0.8036, 0.9643)	0.7954 (0.6286, 0.9268)
ACO	0.6997 (0.5978, 0.7939)	0.7034 (0.6364, 0.7697)	0.7359 (0.5806, 0.8750)	0.6950 (0.6169, 0.7727)	0.4991 (0.3378, 0.6195)	0.6715 (0.4834, 0.8438)	0.6137 (0.5000, 0.7286)	0.6670 (0.4118, 0.9000)	0.5990 (0.4667, 0.7288)	0.3942 (0.1429, 0.5882)
FA	0.6701 (0.5605, 0.7740)	0.7154 (0.6485, 0.7818)	0.7059 (0.5455, 0.8571)	0.7180 (0.6412, 0.7923)	0.4977 (0.3162, 0.6226)	0.8527 (0.7308, 0.9526)	0.7433 (0.6429, 0.8429)	0.7353 (0.5000, 0.9412)	0.7455 (0.6271, 0.8547)	0.5384 (0.2727, 0.7222)
BA	0.8968 (0.8209, 0.9565)	0.8784 (0.8242, 0.9273)	0.8227 (0.6818, 0.9412)	0.8930 (0.8372, 0.9440)	0.7338 (0.6111, 0.8387)	0.8502 (0.7490, 0.9350)	0.8295 (0.7429, 0.9143)	0.8681 (0.6667, 1.0000)	0.8189 (0.7119, 0.9138)	0.6795 (0.4848, 0.8372)
ABC	0.8456 (0.7821, 0.9020)	0.7880 (0.7273, 0.8485)	0.8816 (0.7619, 0.9730)	0.7636 (0.6889, 0.8346)	0.6297 (0.5106, 0.7368)	0.8237 (0.7181, 0.9133)	0.7153 (0.6143, 0.8143)	0.8673 (0.6667, 1.0000)	0.6738 (0.5472, 0.7963)	0.5608 (0.3636, 0.7234)
WOA	0.6503 (0.5327, 0.7597)	0.7332 (0.6667, 0.8000)	0.6169 (0.4483, 0.7813)	0.7635 (0.6899, 0.8333)	0.4689 (0.2645, 0.6139)	0.6952 (0.5395, 0.8367)	0.6851 (0.5714, 0.7857)	0.6668 (0.4000, 0.9091)	0.6900 (0.5660, 0.8103)	0.4500 (0.1736, 0.6500)
Adam	0.9260 (0.8722, 0.9699)	0.9272 (0.8848, 0.9636)	0.7649 (0.6129, 0.9000)	0.9694 (0.9375, 0.9926)	0.8102 (0.6939, 0.9057)	0.6519 (0.4494, 0.8407)	0.8865 (0.8000, 0.9571)	0.4705 (0.2143, 0.7333)	1.0000 (1.0000, 1.0000)	0.6286 (0.3529, 0.8462)
RMSProp	0.9226 (0.8655, 0.9693)	0.9275 (0.8848, 0.9636)	0.7362 (0.5806, 0.8788)	0.9771 (0.9480, 1.0000)	0.8044 (0.6866, 0.9041)	0.6840 (0.8419, 0.9143)	0.8288 (0.7429, 0.9143)	0.4000 (0.1429, 0.6667)	0.9454 (0.8776, 1.0000)	0.4677 (0.1111, 0.7200)
SGD	0.5960 (0.4876, 0.7026)	0.5214 (0.4424, 0.6000)	0.7656 (0.6133, 0.9032)	0.4581 (0.3731, 0.5455)	0.3868 (0.2384, 0.5000)	0.6420 (0.4790, 0.7928)	0.5429 (0.4286, 0.6571)	0.7994 (0.5714, 1.0000)	0.4727 (0.3396, 0.6034)	0.4164 (0.2042, 0.5806)

 https://doi.org/10.1007/s00521-026-12047-6

impact on test-set performance, indicating that baseline initialization and default kinetic formulations were sufficient for RVTE discrimination. Collectively, these results demonstrate that framework-level preprocessing and representation choices, rather than mechanistic parameter variations, are the primary drivers of performance differences across optimization strategies.

The bootstrap DeLong analysis confirmed significant differences in discriminatory performance among optimization strategies (Table 19). The GWO achieved a mean AUC of 0.917 (95% CI: 0.841–0.973), serving as the reference method. Although PSO produced a numerically higher mean AUC (0.940; 95% CI: 0.880–0.984), the difference relative to GWO was not statistically significant (ΔAUC = 0.023, $p = 0.554$), indicating comparable discriminative performance. In contrast, GWO significantly outperformed several metaheuristic optimizers, including the GA ($p = 0.0168$), ACO ($p = 0.0178$), and WOA ($p = 0.0080$). GWO also demonstrated clear statistical superiority over gradient-based optimizers: Adam ($p = 0.0076$), RMSProp ($p = 0.0100$), and SGD ($p = 0.0022$). All gradient-based optimizers showed substantially lower AUCs. Differences between GWO and the FA and BA were not statistically significant, while ABC approached but did not reach significance ($p = 0.0528$). Overall, these results identify GWO and PSO as the top-performing optimizers in terms of AUC, with GWO exhibiting the most consistent and statistically robust discrimination.

The evaluated optimization models displayed distinct and complementary performance profiles regarding calibration, clinical utility, and threshold stability (Fig. 12). In the calibration analysis (Fig. 12A), the GWO- and PSO-optimized frameworks demonstrated the highest fidelity, with predicted risks showing strong agreement with observed event rates across the entire risk spectrum. The calibration curves for these methods tracked the ideal reference line more closely than those of competing metaheuristics and exhibited narrower bootstrap CIs, suggesting enhanced model stability and reduced predictive uncertainty.

The clinical impact was further quantified using DCA (Fig. 12B), underscoring the utility of GWO and PSO. Across a broad range of clinically relevant threshold probabilities, both methods consistently achieved a higher net benefit than alternative optimizers or "treat-all/treat-none" strategies. Notably, GWO maintained the peak net benefit across most thresholds, representing an optimal trade-off between sensitivity for RVTE events and minimizing unnecessary clinical interventions.

Finally, threshold stability was assessed using Youden's index plateaus on the ETP scale (Fig. 12C). GWO and PSO yielded balanced sensitivity and specificity without relying on extreme or unstable cutoffs. The broad plateau observed for these optimizers, as opposed to the sharp, localized peaks seen in other methods, suggests a robust decision-making zone. This threshold behavior is particularly critical for clinical translation, as it ensures that the model remains reliable even under slight variations in population-specific cutoffs. Collectively, these findings position GWO and PSO as the most robust candidates for threshold-based clinical decision support in RVTE risk stratification.

To investigate how systemic clinical profiles relate to localized biochemical mechanisms and how these relationships vary with recurrence status, we performed a stratified Spearman correlation analysis using the optimal ANN–GWO and ANN–PSO surrogate frameworks. Results are summarized in Figs. 13, 14, 15 and 16, and provide complementary insight into both stable and RVTE-specific clinical–mechanistic couplings.

The stratified rank correlation maps for RVTE− and RVTE+ patients (Figs. 13 and 15) reveal a heterogeneous pattern of predominantly weak-to-moderate associations across kinetic parameters and clinical variables. This diffuse structure indicates that coagulation kinetics are governed by nonlinear, multifactorial interactions rather than by a single dominant clinical driver.

Despite this complexity, several hematological indices exhibited consistent and recurrent associations across both recurrence strata and both optimization frameworks. In particular, red blood cell (RBC) count, hemoglobin, and hematocrit showed the most robust and spatially coherent correlations with multiple kinetic parameters, suggesting a stable link between erythrocyte-related properties and coagulation dynamics. White blood cell (WBC) count and antithrombin III (AT-III) emerged as secondary but statistically significant modulators, consistent with the contribution of inflammatory and anticoagulant pathways to thrombin generation.

Table 17 Performance of ODE-based reference models optimized by metaheuristic and gradient-based algorithms (bootstrap median and 95% CIs for training and testing metrics)

MOA	AUC Train	ACC Train	TPR Train	TNR Train	F1score Train	AUC Test	ACC Test	TPR Test	TNR Test	F1Score Test
GA	0.7619 (0.6841, 0.8343)	0.7216 (0.6545, 0.7879)	0.9419 (0.8529, 1.0000)	0.6645 (0.5814, 0.7443)	0.5801 (0.4632, 0.6838)	0.6801 (0.5463, 0.8023)	0.6300 (0.5143, 0.7429)	0.7985 (0.5714, 1.0000)	0.5838 (0.4528, 0.7143)	0.4699 (0.2424, 0.6414)
PSO	0.8486 (0.7433, 0.9357)	0.9152 (0.8727, 0.9576)	0.8820 (0.7586, 0.9730)	0.9239 (0.8769, 0.9645)	0.8086 (0.6984, 0.8986)	0.9239 (0.8551, 0.9769)	0.8285 (0.7429, 0.9143)	0.8660 (0.6667, 1.0000)	0.8182 (0.7143, 0.9138)	0.6776 (0.4848, 0.8372)
GWO	0.9354 (0.8711, 0.9843)	0.9455 (0.9091, 0.9758)	0.9413 (0.8485, 1.0000)	0.9466 (0.9051, 0.9837)	0.8749 (0.7813, 0.9500)	0.8942 (0.8071, 0.9636)	0.8999 (0.8286, 0.9571)	0.9343 (0.7778, 1.0000)	0.8905 (0.8000, 0.9643)	0.7953 (0.6250, 0.9231)
ACO	0.7408 (0.6487, 0.8274)	0.6971 (0.6242, 0.7636)	0.7351 (0.5833, 0.8788)	0.6872 (0.6064, 0.7647)	0.4931 (0.3279, 0.6111)	0.5461 (0.3850, 0.7043)	0.5863 (0.4714, 0.7000)	0.6661 (0.4000, 0.9000)	0.5643 (0.4333, 0.6923)	0.3768 (0.1326, 0.5714)
FA	0.6855 (0.5776, 0.7927)	0.7087 (0.6364, 0.7758)	0.7049 (0.5484, 0.8571)	0.7098 (0.6290, 0.7865)	0.4907 (0.3210, 0.6154)	0.8308 (0.7060, 0.9350)	0.7580 (0.6571, 0.8571)	0.8009 (0.5714, 1.0000)	0.7462 (0.6250, 0.8571)	0.5780 (0.3429, 0.7500)
BA	0.8858 (0.8085, 0.9469)	0.8420 (0.7818, 0.8970)	0.8522 (0.7222, 0.9667)	0.8393 (0.7731, 0.9008)	0.6874 (0.5641, 0.7957)	0.8486 (0.7492, 0.9349)	0.7288 (0.6143, 0.8286)	0.9340 (0.7778, 1.0000)	0.6730 (0.5472, 0.7931)	0.5907 (0.4087, 0.7500)
ABC	0.8374 (0.7748, 0.8940)	0.7815 (0.7212, 0.8424)	0.8825 (0.7586, 0.9737)	0.7553 (0.6818, 0.8277)	0.6221 (0.5000, 0.7306)	0.8176 (0.7149, 0.9085)	0.7151 (0.6000, 0.8143)	0.8677 (0.6667, 1.0000)	0.6732 (0.5470, 0.7925)	0.5604 (0.3615, 0.7200)
WOA	0.6369 (0.5224, 0.7469)	0.7337 (0.6667, 0.8000)	0.6188 (0.4483, 0.7826)	0.7635 (0.6884, 0.8333)	0.4693 (0.2608, 0.6108)	0.6484 (0.4984, 0.7895)	0.6870 (0.5714, 0.7857)	0.6670 (0.4167, 0.9091)	0.6925 (0.5636, 0.8077)	0.4523 (0.1832, 0.6512)
Adam	0.9049 (0.8443, 0.9543)	0.8181 (0.7576, 0.8727)	0.8826 (0.7647, 0.9730)	0.8014 (0.7308, 0.8682)	0.6641 (0.5430, 0.7692)	0.6145 (0.4112, 0.8057)	0.7568 (0.6571, 0.8571)	0.4643 (0.2000, 0.7273)	0.8367 (0.7347, 0.9273)	0.3955 (0.0833, 0.6471)
RMSProp	0.8991 (0.8358, 0.9520)	0.8061 (0.7455, 0.9520)	0.8829 (0.7619, 0.9730)	0.7861 (0.7132, 0.8538)	0.6498 (0.5263, 0.7600)	0.6382 (0.4570, 0.8053)	0.7145 (0.6000, 0.8143)	0.4649 (0.2000, 0.7273)	0.7826 (0.6667, 0.8868)	0.3456 (0.0672, 0.6087)
SGD	0.5949 (0.4884, 0.6963)	0.5390 (0.4606, 0.6121)	0.7352 (0.5806, 0.8800)	0.4881 (0.4015, 0.5736)	0.3816 (0.2286, 0.5000)	0.6188 (0.4406, 0.7845)	0.5570 (0.4429, 0.6714)	0.7339 (0.5000, 0.9375)	0.5088 (0.3750, 0.6364)	0.3938 (0.1645, 0.5714)

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Significant correlations were sparsely distributed, reinforcing that no single clinical variable uniformly governs the mechanistic landscape. Instead, the observed associations reflect distributed sensitivity across multiple physiological domains.

To explicitly quantify recurrence-dependent shifts in clinical–mechanistic coupling, we computed Δρ maps (Figs. 14 and 16), defined as the difference in Spearman correlation coefficients between the RVTE + and RVTE− groups. While the majority of Δρ values clustered near zero, indicating broadly conserved association patterns, a subset of kinetic parameters exhibited statistically significant deviations, highlighting RVTE-specific modulation.

Positive Δρ values were most frequently associated with erythrocyte-related indices (RBC count, hemoglobin, and hematocrit), indicating that, in RVTE+ patients, variations in these variables exert a disproportionately stronger influence on kinetic rates than in the non-recurrent group. This pattern may reflect altered blood rheology, erythrocyte–leukocyte interactions, or potential alterations in erythrocyte-mediated microenvironmental effects in patients prone to recurrence.

A pronounced and consistent finding across both frameworks was the age-dependent amplification of associations with $Kcat_{47}$ and $Kcat_{14}$, corresponding to the activation of coagulation factors X and V, respectively. These parameters are central components of the prothrombinase complex, and their age-sensitive modulation in RVTE+ patients suggests selective sensitization of key amplification pathways within the coagulation cascade.

In addition, the duration of anticoagulation therapy showed positive Δρ values across several kinetic parameters, suggesting that long-term treatment may not only suppress coagulation activity but also be associated with altered coupling patterns, particularly in patients with recurrent disease. WBC count and AT-III also demonstrated localized but significant Δρ shifts, consistent with recurrence-specific inflammatory and anticoagulant adaptations.

Importantly, these RVTE-associated changes were distributed across multiple clinical domains rather than driven by a single factor, underscoring the multifactorial architecture of recurrence risk.

Comparison of the GWO- and PSO-optimized surrogates demonstrates strong concordance in the overall topology of the stratified correlation and Δρ maps. Both optimizers consistently identified the dominance of the erythrocyte-related triad (RBC, hemoglobin, hematocrit) and the age-dependent sensitivity of factor X and factor V activation, supporting the biological robustness of these findings.

Subtle differences in sensitivity were observed. The PSO-derived Δρ maps (Fig. 16) exhibited slightly enhanced detection of significant shifts involving AT-III and WBC count, suggesting increased sensitivity to lower-magnitude inflammatory or anticoagulant effects. In contrast, the GWO framework yielded a marginally stronger and more spatially coherent signal for age-related modulation of kinetic parameters. Nevertheless, these differences were quantitative rather than qualitative.

The convergence of two distinct heuristic optimization strategies on the same core biological signatures provides internal validation of the surrogate modeling framework. It supports the interpretation that the observed associations reflect intrinsic properties of the patient cohorts rather than optimizer-specific artifacts.

The transition from an RVTE−to an RVTE+ phenotype is characterized by a distributed reweighting of clinical influence on coagulation kinetics rather than by wholesale reorganization of the association structure. Age-dependent modulation of factor X and V activation and the amplified role of hematological indices in recurrent patients define a distinct biochemical signature of recurrence. These findings support a mechanistically grounded, multifactorial view of RVTE risk and highlight the potential of clinical–mechanistic coupling analyses for improved, personalized risk stratification in VTE. It is important to note that these associations should be interpreted with caution. Although the observed relationships are internally consistent, they arise from a pseudo-supervised model-derived surrogate framework rather than from direct experimental supervision. Consequently, external validation and complementary experimental approaches—such as integration with thrombin generation assays—are necessary to strengthen the robustness, biological plausibility, and scientific credibility of the reported associations.

The normalized permutation-importance matrices were visualized as heatmaps (Figs. 17 and 18), providing a global, parameter-specific view of clinical feature sensitivity within the learned surrogate models. This analysis

Table 18 Sensitivity of RVTE prediction performance to framework configuration across metaheuristic optimizers

Metaheuristic Optimizer	Framework Component	Best-Performing Configuration	ΔAUC (Best − Worst)	Partial R^2 (Drop-One)	*p*-value	Best Mean Test AUC
GA	Normalization	Standardized	0.058	0.182	< 0.001	0.703
	Selected Features	All features	0.033	0.070	< 0.001	0.691
PSO	Normalization	Standardized	0.060	0.163	< 0.001	0.841
	Kinetic Parameters	Full parameter set	0.032	0.052	0.002	0.827
GWO	Normalization	Standardized	0.075	0.289	< 0.001	0.829
	Hidden Neurons	50	0.045	0.064	< 0.001	0.808
	Selected Features	All features	0.019	0.027	0.025	0.801
ACO	Selected Features	PCA-based	0.040	0.074	< 0.001	0.612
FA	Normalization	Standardized	0.028	0.037	0.009	0.653
	Selected Features	PCA-based	0.027	0.035	0.010	0.652
BA	Normalization	Standardized	0.052	0.158	< 0.001	0.791
ABC	Fitness Score	DMP-L2 (full regularization)	0.052	0.096	< 0.001	0.750
	Normalization	Standardized	0.029	0.046	0.003	0.737
	Selected Features	PCA-based	0.021	0.026	0.028	0.733
WOA	Hidden Neurons	50	0.033	0.022	0.045	0.630

Notes: ΔAUC represents the difference between the highest and lowest mean test AUC observed across levels of each framework component. Partial R^2 was computed using a drop-one model strategy and reflects the proportion of variance explained by each setting relative to the unexplained variance. Only statistically significant factors ($p < 0.05$) are reported. Reported AUC values correspond to the mean performance across configurations that share the specified setting and do not necessarily reflect the single globally optimal model

Table 19 Pairwise DeLong comparison of AUCs across optimization algorithms for RVTE prediction

Optimizer	Mean AUC	95% CI (Lower)	95% CI (Upper)	ΔAUC vs. GWO	Z-Statistic	*p*-Value
GWO	0.9168	0.8406	0.9733	-	-	-
GA	0.7677	0.6378	0.8800	− 0.1491	2.1199	0.0168
PSO	0.9397	0.8804	0.9842	0.0230	− 0.4870	0.5540
ACO	0.6722	0.4788	0.8462	− 0.2446	2.3591	0.0178
FA	0.8523	0.7327	0.9492	− 0.0645	0.9382	0.3074
BA	0.8500	0.7512	0.9350	− 0.0668	1.0724	0.2410
ABC	0.8240	0.7207	0.9109	− 0.0928	1.6329	0.0528
WOA	0.6950	0.5397	0.8364	− 0.2218	2.5304	0.0080
Adam	0.6510	0.4481	0.8358	− 0.2658	2.6266	0.0076
RMSProp	0.6842	0.5100	0.8434	− 0.2326	2.4092	0.0100
SGD	0.6432	0.4791	0.7942	− 0.2735	3.1421	0.0022

quantifies how perturbations in individual clinical inputs propagate through the ANN to influence distinct kinetic parameters of the coagulation cascade.

Permutation analysis of the ANN–GWO model (Fig. 17) revealed pronounced heterogeneity in feature importance across kinetic parameters, indicating that distinct subsets of clinical variables govern different biochemical reactions. No single input emerged as a universal driver; instead, importance patterns were highly structured and parameter-specific, consistent with the multifactorial nature of coagulation kinetics.

Across kinetic parameters, hematologic variables dominated the importance landscape. Red blood cell–related indices, such as RBC count, hemoglobin, and hematocrit, formed a coherent cluster of high importance and acted as primary drivers for multiple activation and saturation parameters, including $Kcat_9$, $Kcat_{12}$, $Kcat_{14}$, Km_{15}, and K_{50}. This finding supports a central role for erythrocyte-related properties in shaping reaction rates and effective enzyme–substrate interactions within the coagulation system.

WBC count emerged as the single most influential feature on average in the GWO-optimized model, exhibiting moderate-to-high importance across a broad set of kinetic parameters. This diffuse influence suggests that

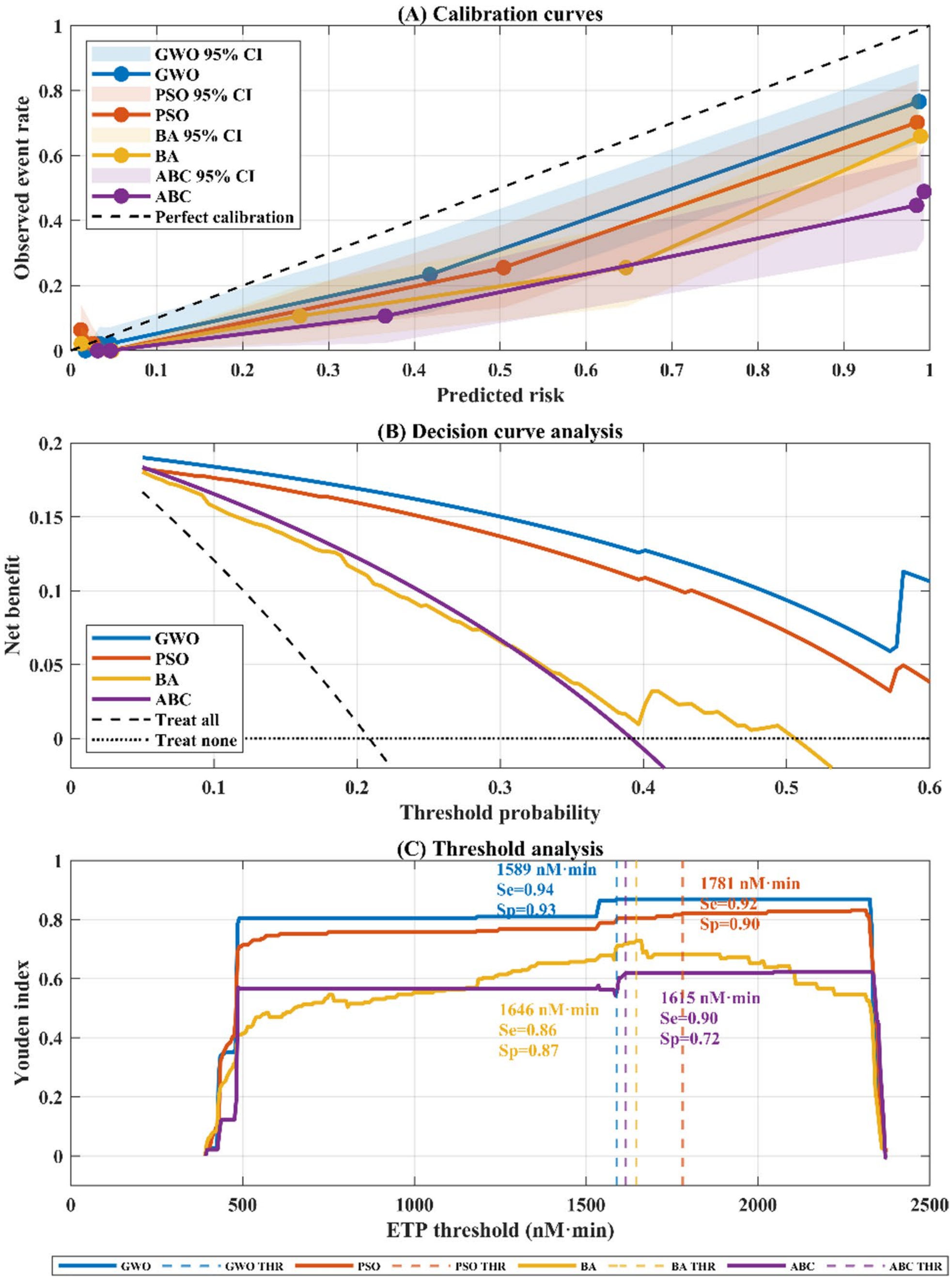

Fig. 12 Comparative evaluation of (**A**) calibration curves, (**B**) clinical utility, and (**C**) decision stability across MOAs. THR: Threshold. Se: sensivity. Sp: Specificity

inflammatory or immune-related processes act as global modulators rather than targeting a single reaction step, consistent with the known crosstalk between leukocytes and thrombin generation.

Age also displayed strong and selective importance, particularly for activation constants such as $Kcat_{10}$, $Kcat_{11}$, and $Kcat_{13}$, as well as downstream parameters including K_{49}. These patterns reinforce the role of age as a modulator of amplification dynamics rather than baseline reaction rates.

In contrast, lipid-related variables and metabolic markers generally showed lower and more localized importance, with HDL cholesterol and glucose exerting sparse but non-negligible influence on specific parameters such as K_{17} and K_{53}. Collectively, these results underscore that the ANN–GWO model's predictive capacity arises from integrating multiple clinical domains rather than relying on a narrow subset of inputs.

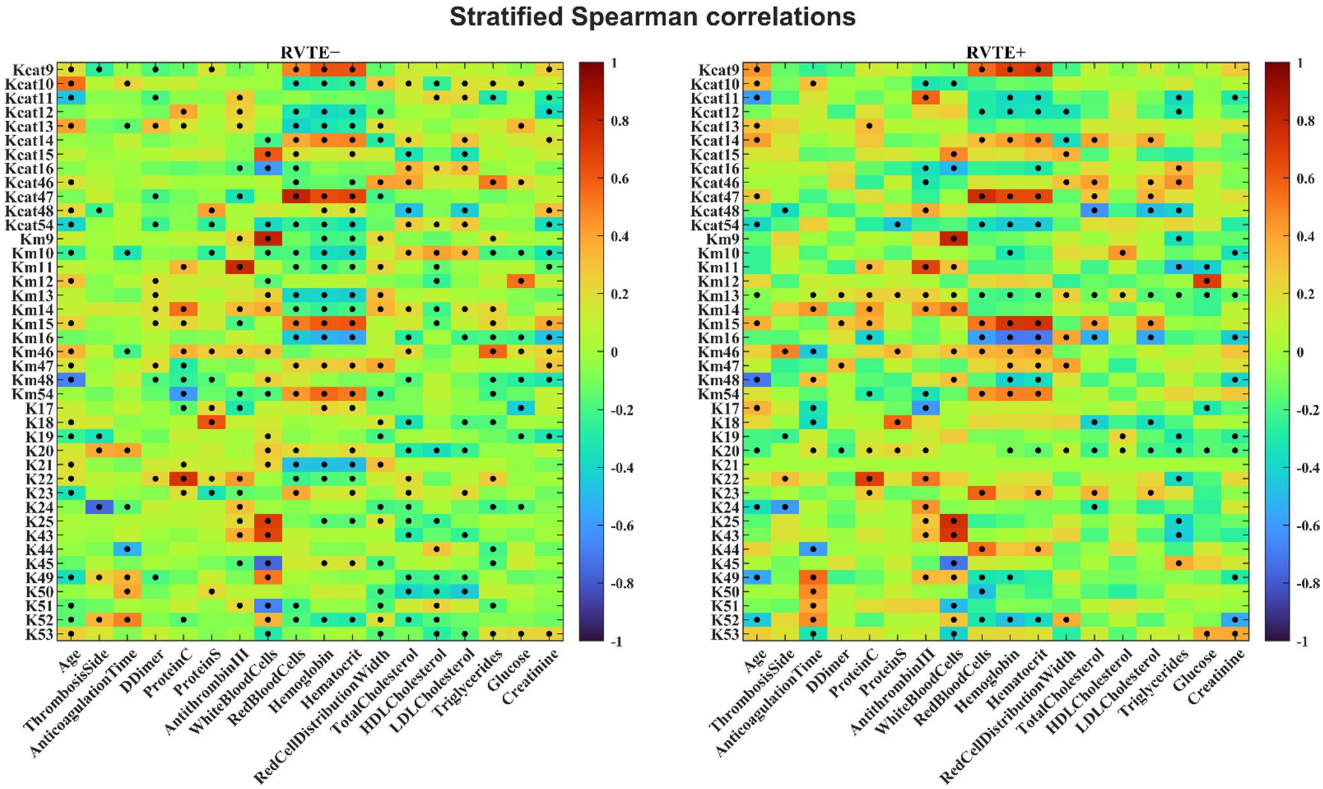

Fig. 13 Stratified association analysis of clinical inputs and mechanistic kinetic parameters. Heatmaps display Spearman rank correlation coefficients (ρ) for the RVTE − and RVTE+ subgroups, generated using the optimal ANN-GWO framework. Significant associations, defined as those where the 95% bootstrap CI (10,000 iterations) excludes zero, are indicated by black markers

Entropy-based analysis further demonstrated that feature importance in the GWO model was relatively concentrated (mean entropy = 3.84), indicating sharper reliance on dominant hematologic and inflammatory features.

Permutation analysis of the ANN–PSO model (Fig. 18) revealed a broadly consistent yet quantitatively distinct importance structure. While hematologic variables remained influential, the PSO-optimized network distributed importance more evenly across clinical inputs.

Age emerged as the most influential feature overall in the PSO model, followed closely by thrombosis side, total cholesterol, and LDL cholesterol, indicating a stronger integration of demographic and metabolic information compared with the GWO surrogate. Several kinetic parameters, including K_{50} and K_{51}, consistently retained sensitivity to red blood cell indices across both optimizers, suggesting robust erythrocyte-driven effects independent of the training heuristic.

Notably, Protein C ranked among the top contributors in both GWO and PSO frameworks, highlighting its stable mechanistic relevance. Protein S exerted a selective effect on K_{18} in both models, further supporting biological plausibility.

Compared with GWO, PSO exhibited a significantly higher entropy of importance (mean entropy = 5.30), indicating a more diffuse sensitivity profile. This redistribution is consistent with broader integration of the multidimensional input space during PSO optimization, reducing dominance by individual features.

Direct comparison of permutation-importance matrices revealed minimal global agreement between GWO and PSO at the individual-entry level (Spearman's $\rho = 0.03$, $p = 0.42$), indicating that local feature–parameter sensitivities differ substantially between optimization strategies. However, this apparent discrepancy masks strong agreement at a higher structural level.

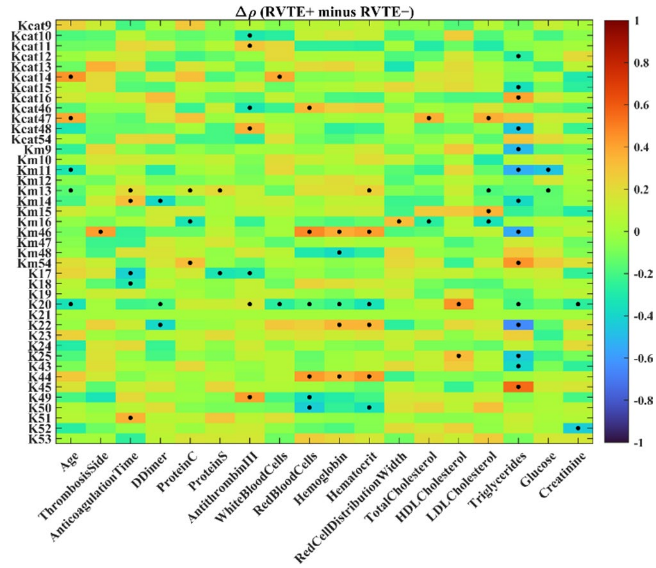

Fig. 14 Δρ analysis of clinical inputs and mechanistic kinetic parameters, generated using the optimal ANN-GWO framework. Visualization of the change in correlation strength between RVTE + and RVTE− strata. Significant associations, defined as those where the 95% bootstrap CI (10,000 iterations) excludes zero, are indicated by black markers

Both optimizers consistently identified hematologic indices and age as dominant drivers, with age and protein C emerging as shared top-ranked features. Moreover, several kinetic parameters, particularly $Kcat_{14}$, $Kcat_{15}$, $Kcat_{47}$, Km_9, Km_{12}, and K_{43}, exhibited large optimizer-dependent shifts in sensitivity, suggesting that these reactions are especially responsive to alternative parameterizations of the surrogate model.

Overall, the permutation analysis demonstrates that while GWO and PSO differ in how they distribute importance across inputs, they converge on the same core biological domains. GWO yields a more concentrated, feature-selective importance structure, whereas PSO promotes broader, more balanced integration of clinical information.

To evaluate robustness to stochastic variability, each metaheuristic was executed over 30 independent runs using fixed algorithmic settings from the top-performing models and distinct random initializations. Summary statistics of final fitness, convergence efficiency, and predictive performance are reported in Table 20.

Across repeated runs, GWO and PSO consistently emerged as the two best-performing algorithms, clearly outperforming the remaining metaheuristics in terms of optimization quality and discriminative performance. Both methods achieved nearly identical mean final fitness values (PSO: 238,477; GWO: 236,845) and comparable best-case solutions exceeding 132,800, indicating a similar capacity to explore and exploit high-quality regions of the search space. In contrast, GA, FA, ABC, BA, ACO, and WOA exhibited markedly lower fitness values and greater variability, reflecting reduced robustness to stochastic initialization.

While the final optimization outcomes of GWO and PSO were highly comparable, differences were primarily observed in convergence dynamics rather than solution quality. PSO reached 90% of its final fitness on average earlier (61.5% of the optimization horizon) than GWO (42.2%), suggesting faster convergence. However, GWO demonstrated competitive convergence behavior with acceptable variability across runs, indicating stable albeit slower optimization trajectories.

In terms of predictive performance, both GWO and PSO achieved the highest bootstrap AUC values on the training and test sets, with only marginal differences between them (PSO: 0.881 train, 0.849 test; GWO: 0.875 train, 0.821 test). These small performance gaps were accompanied by overlapping variability ranges, supporting the conclusion that both algorithms offer comparable generalization capability. The remaining metaheuristics produced substantially lower AUC values and higher dispersion, particularly on the test set, highlighting inferior stability under stochastic perturbations.

Statistical analysis corroborated these observations. The Friedman test ranked GWO and PSO as the top-performing methods. Subsequent Nemenyi post-hoc comparisons revealed no statistically significant difference between GWO and PSO. GWO significantly outperformed all remaining algorithms, whereas PSO significantly

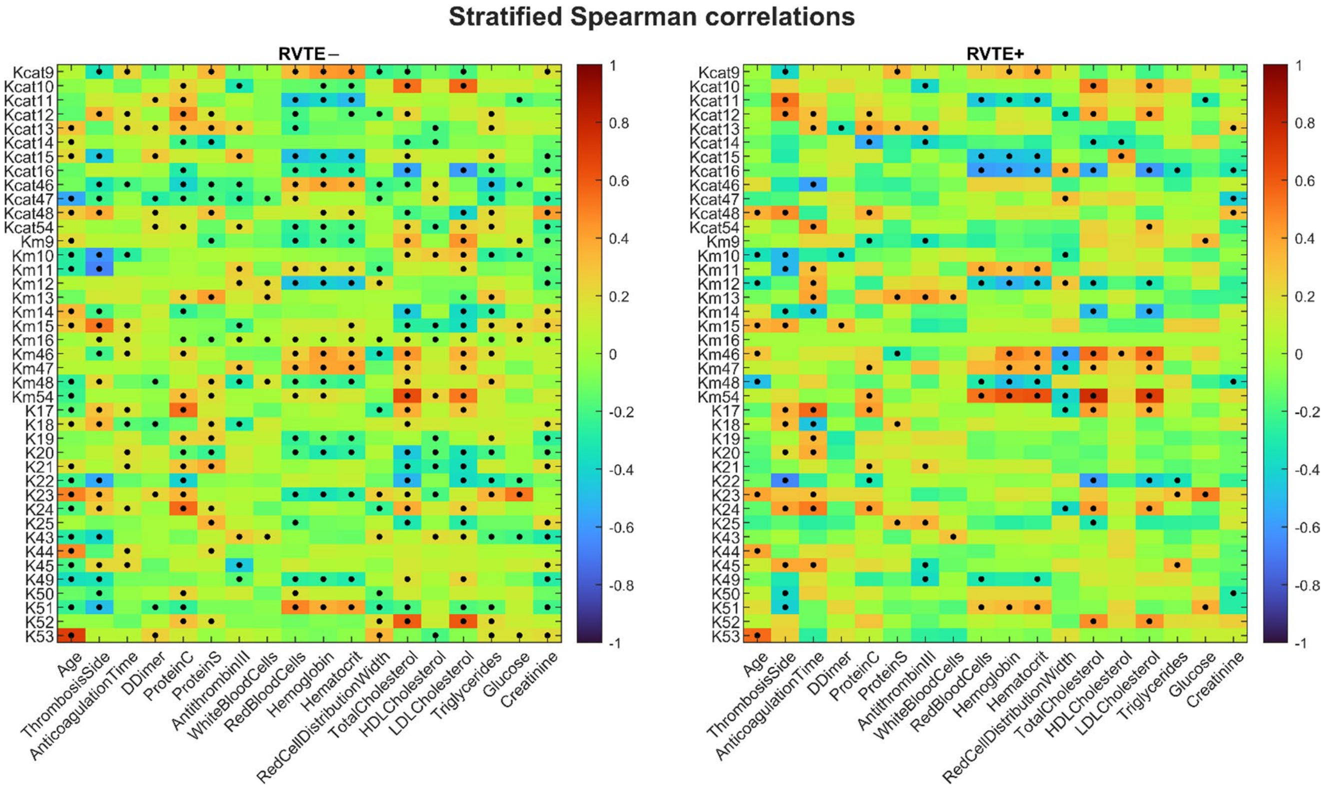

Fig. 15 Stratified association analysis of clinical inputs and mechanistic kinetic parameters. Heatmaps display Spearman rank correlation coefficients (ρ) for the RVTE−and RVTE+ subgroups, generated using the optimal ANN-PSO framework. Significant associations, defined as those where the 95% bootstrap CI (10,000 iterations) excludes zero, are indicated by black markers

outperformed all others except BA, for which the difference did not reach statistical significance. Taken together, these findings indicate that GWO and PSO provide similarly robust optimization performance, with PSO offering faster convergence and GWO representing a competitive alternative with comparable final solution quality and generalization performance.

4 Discussion

The present study demonstrates that replacing a computationally intensive ODE-based coagulation model with an ANN surrogate enables substantial gains in efficiency without compromising predictive fidelity. Linear regression methods confirmed the highly nonlinear structure of the problem, as they yielded relatively high and comparable errors. In contrast, boosting algorithms and ANN achieved markedly improved fits, consistent with their established ability to model nonlinear interactions in different applications [61, 62]. The ANN achieved the lowest prediction errors (RMSE < 0.3 in both the training and test sets), highlighting its suitability as a surrogate for complex biochemical dynamics.

The best-performing ANN comprised 385 adjustable parameters compared with 41 kinetic parameters in the mechanistic ODE system. Despite the larger parameter space, the ANN achieved a drastic computational advantage, reducing runtime by more than 99% while maintaining high fidelity, with a relative accuracy of 97.97% under ANN-GWO optimization. A similar example of reducing an intensive optimization problem through model replacement with an ANN is reported in the literature [63], where substituting a computationally demanding ODE system with an ANN during multiple optimization steps (as in model predictive control) decreased the

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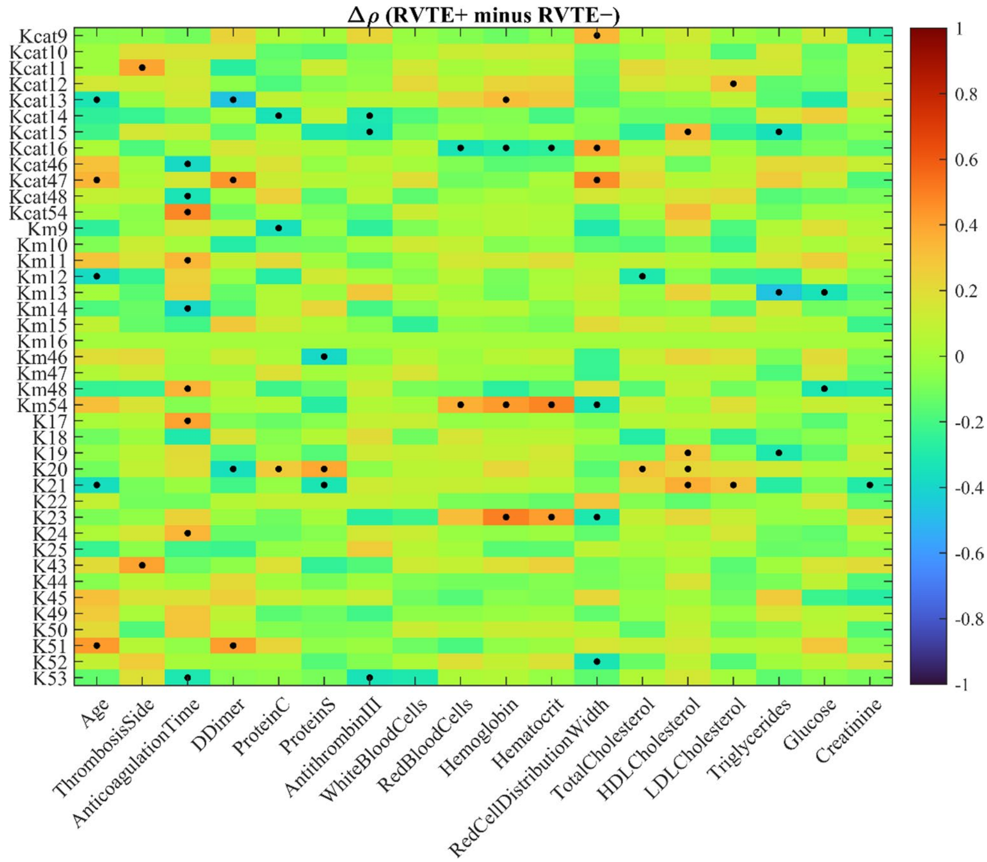

Fig. 16 Δρ analysis of clinical inputs and mechanistic kinetic parameters, generated using the optimal ANN-PSO framework. Visualization of the change in correlation strength between RVTE + and RVTE− strata. Significant associations, defined as those where the 95% bootstrap CI (10,000 iterations) excludes zero, are indicated by black markers

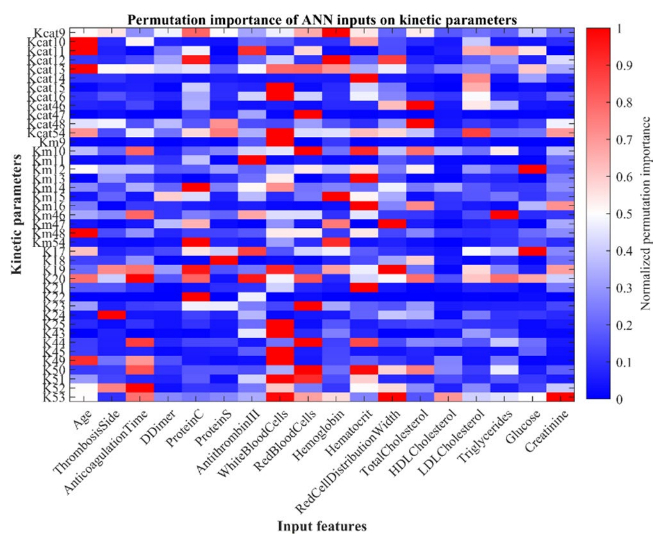

Fig. 17 Relative feature importance for clinical-to-kinetic parameter mapping. Heatmap of normalized permutation importance values, highlighting the dependence of the ANN-GWO on specific hematologic, metabolic, and clinical inputs

computational cost by approximately three orders of magnitude (around 1000-fold). This efficiency gain is particularly critical in hybrid mechanistic–data-driven frameworks, where repeated model evaluations constitute the principal computational bottleneck.

Within the hybrid framework, z-score normalization consistently emerged as the most effective preprocessing strategy, suggesting its role as a practical default for similar applications. This pattern aligns with the findings of

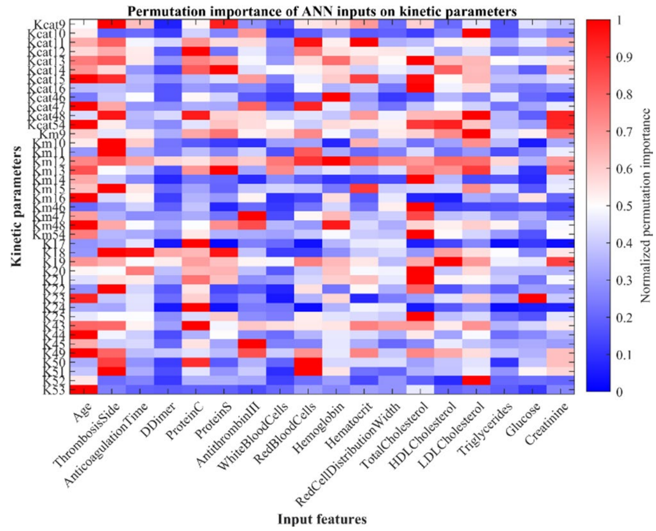

Fig. 18 Relative feature importance for clinical-to-kinetic parameter mapping. Heatmap of normalized permutation importance values, highlighting the dependence of the ANN-PSO on specific hematologic, metabolic, and clinical inputs

Table 20 Robustness, convergence efficiency, predictive performance, and statistical comparison of metaheuristic algorithms over 30 independent runs with ANN surrogate

Metric	Statistics	GA	PSO	GWO	ACO	FA	BA	ABC	WOA
Final DMP	Worst	95,112.75	190,068.89	204,295.83	39,579.67	79,653.98	160,937.75	124,601.29	1,590.74
	Mean	117,737.58	238,477.51	236,845.91	56,120.81	88,733.67	201,625.39	161,295.02	10,427.98
	Best	139,147.39	265,960,50	265,197.45	68,258.74	99,489.55	236,029.11	203,229.47	47,628.35
	SD	6,970.24	19,856.29	16,620.44	48,406.41	4,922.55	16,411.60	19,210.44	11,252.15
Convergence Efficiency (%)	Worst	43.00	42.67	26.33	99.67	48.00	6.67	32.67	81.33
	Mean	76.92	61.46	42.22	99.67	78.84	59.90	55.00	97.98
	Best	95.00	77.00	74.67	99.67	95.33	89.33	82.33	99.67
	SD	13.47	8.88	10.80	0.00	12.85	22.89	13.36	3.65
Bootstrap AUC (Train)	Worst	0.6300	0.7973	0.8139	0.5418	0.5864	0.7417	0.6893	0.3266
	Mean	0.7494	0.8811	0.8749	0.6355	0.6912	0.8542	0.7808	0.4726
	Best	0.8227	0.9337	0.9310	0.7107	0.7498	0.9280	0.8699	0.6012
	SD	0.0387	0.0335	0.0354	0.0424	0.0399	0.0494	0.0469	0.0572
Bootstrap AUC (Test)	Worst	0.5492	0.7045	0.6978	0.4214	0.4841	0.6477	0.5731	0.3180
	Mean	0.6995	0.8487	0.8213	0.5770	0.6432	0.7955	0.7354	0.4852
	Best	0.8163	0.9553	0.9250	0.7346	0.7850	0.9390	0.8622	0.5886
	SD	0.0695	0.0622	0.0455	0.0774	0.0889	0.0632	0.0630	0.0716
Friedman test	Rank	5	2	1	7	6	3	4	8
	Average Rank	5.00	1.77	1.23	6.83	6.17	3.20	3.80	8.00
Nemenyi post-hoc	p-value vs. GWO	<0.001	0.991	-	<0.001	<0.001	0.039	0.001	<0.001
	Significance	Very significant	Not significant	-	Very significant	Very significant	Significant	Significant	Very significant
	p-value vs. PSO	<0.001	-	0.991	<0.001	<0.001	0.312	0.029	<0.001
	Significance	Very significant	-	Not significant	Very significant	Very significant	Not significant	Significant	Very significant

the study by Singh and Singh [64], which reported that the z-score normalization method achieved the best performance across 20 medical datasets using ML. The optimal configuration also employed initial concentrations of coagulation species reported by Zhu [35], varying only the kinetic parameters during training. This strategy is clinically relevant, as molar concentrations of most coagulation factors are rarely available in practice. Additionally, feature reduction via PCA was often beneficial, lowering data collection demands. However, this came at the cost of reduced interpretability, as dimensionality reduction can obscure the physiological role of individual variables in thrombin generation and RVTE prediction.

The ANN-GWO model achieved strong classification performance, with AUC = 0.89455, TPR = 0.933, and TNR = 0.89 on the independent test set. Reported AUC values in the literature vary widely, from moderate (AUC = 0.69 with DT [65]) to near-perfect (AUC = 0.9973 with ANN [23]). More recent ANN-based approaches typically report AUC values above 0.85 [23, 36], often supported by dimensionality reduction methods such as PCA or PLS. Although the ANN-GWO achieved a slightly lower AUC, it provided a balanced trade-off between sensitivity and specificity, in contrast to the inflated TPR = 1.0 commonly reported in prior studies [20, 23, 36], which may suggest overfitting or limited generalizability. Unlike purely data-driven models, the proposed framework integrates mechanistic surrogate modeling with metaheuristic optimization, enhancing interpretability and translational relevance.

Beyond discrimination, calibration analysis, decision curve analysis, and threshold stability analyses collectively demonstrated the clinical robustness of swarm-based optimizers. GWO and PSO showed superior agreement between predicted and observed risks, consistently higher net benefit across clinically relevant thresholds, and broad Youden index plateaus on the ETP scale. This plateau behavior is particularly relevant for real-world deployment, as it reduces sensitivity to minor threshold perturbations and enhances decision stability. Competing optimizers exhibited narrower stability regions or reduced net benefit, limiting their clinical reliability.

Stratified correlation and $\Delta\rho$ analyses indicated that RVTE recurrence is characterized not by a structural reorganization of clinical–kinetic relationships, but rather by a selective reweighting of existing couplings. Erythrocyte-related indices (RBC count, hemoglobin, and hematocrit) consistently emerged as dominant modulators across recurrence strata and optimization frameworks, reinforcing their systemic influence on coagulation dynamics. In addition, age-dependent amplification of factor X and factor V activation suggests sensitization of key nodes within the prothrombinase complex in recurrent patients. This age-related kinetic reinforcement is biologically plausible and consistent with established hemostatic changes observed in aging populations. Plasma concentrations of several coagulation factors—especially factors V and X—have been shown to increase with age, accompanied by enhanced thrombin generation and platelet activation, collectively contributing to a prothrombotic phenotype [66, 67]. Moreover, population-based studies demonstrate that biomarkers reflecting factor Xa activity and prothrombin conversion, such as prothrombin fragment F1 + 2, rise progressively with age, indicating increased basal thrombin generation in older individuals [68]. The convergence between these clinical observations and the model-derived amplification of kinetic parameters for factors V and X supports the mechanistic plausibility of our findings.

Permutation-importance analysis reinforced this interpretation. Although GWO and PSO distributed feature importance differently at the local level, both consistently identified hematologic variables, age, and Protein C as dominant biological drivers. GWO exhibited a more concentrated importance structure, whereas PSO distributed sensitivity more diffusely across clinical inputs. Despite low agreement at the entry level, both optimizers converged on shared physiological domains.

Across all comparative experiments, GWO and PSO consistently achieved superior performance in final fitness, discrimination, calibration, and robustness. Their superiority persisted across 30 independent stochastic runs, confirming that results were not attributable to favorable initialization. Friedman and Nemenyi tests demonstrated that both methods significantly outperformed most alternative metaheuristics, with no statistically significant difference between GWO and PSO. Although PSO converged more rapidly, both algorithms achieved statistically indistinguishable final solution quality and AUC values. These findings suggest that the surrogate objective landscape remains highly nonconvex and multimodal, favoring global search strategies. Notably, even after replacing the ODE system with a differentiable ANN surrogate, gradient-based optimizers (Adam, RMSProp, SGD) failed to match swarm-based performance. This underscores that differentiability alone does not guarantee effective optimization in complex biological systems characterized by rugged loss surfaces and multiple local minima. These observations are consistent with the recent literature, which shows that swarm-based metaheuristic algorithms can outperform traditional gradient-based methods in training ANNs and other high-dimensional, nonlinear problems due to their enhanced capacity to avoid local minima and to explore rugged, multimodal search spaces through demographic and adaptive search mechanisms [69]. Additionally, systematic reviews of optimization methods in ML underscore that population-based, gradient-free strategies such as PSO and GWO are particularly effective in complex nonconvex landscapes where gradient descent techniques may struggle to escape local optima or to provide reliable global search direction [70, 71].

Direct comparison with alternative surrogate models confirmed the superiority of ANN-GWO over all MOA-based approaches tested with GA, ACO, FA, BA, and WOA, except for CatBoost-BA. ANN-GWO also showed statistically comparable performance to surrogate models optimized with PSO, suggesting that GWO and PSO are particularly effective metaheuristics in this context. No statistically significant improvements were observed when comparing GWO to other ML algorithms. These findings provide practical guidance on the application of GWO and PSO as robust optimizers, and of ANN and boosting-based ML methods as efficient surrogates to ODE systems in hybrid frameworks.

The proposed approach offers substantial practical potential for advancing hybrid mechanistic–data-driven modeling in clinical settings. The original ANN–ODE framework [23] was limited by high computational costs, restricting the exploration of features and kinetic parameter selection. Replacing the ODE with ANN-GWO reduced computation time by over 99%, effectively removing this barrier. This efficiency gain enables the use of refined selection strategies, such as backward and forward elimination, to identify clinical features and kinetic parameters that maintain accuracy while reducing complexity and enhancing interpretability. Moreover, the framework facilitates investigation of how clinical variables (e.g., age, comorbidities, hemostatic biomarkers) qualitatively and quantitatively influence kinetic parameters, and how these, in turn, affect thrombin generation, ETP, and ultimately RVTE risk.

Several limitations should be acknowledged. First, only a single ODE-based coagulation model was examined. Given structural and kinetic differences across alternative coagulation models, such as models of Hockin et al. [72] and Jones and Mann [73], it remains uncertain whether surrogate replacement generalizes universally. Future work should evaluate robustness across multiple mechanistic frameworks.

Second, MOA hyperparameters were adopted from commonly used literature settings rather than systematically tuned. Because hyperparameter configurations strongly influence convergence dynamics and solution quality [74, 75], future studies should perform systematic sensitivity analyses to provide a more rigorous comparison of algorithmic robustness.

Third, the surrogate was designed to predict a single scalar summary of thrombin generation (ETP) rather than the full time-resolved thrombin concentration curve. ETP is clinically interpretable and mechanistically grounded as a global measure of thrombin-generating capacity. It reflects the integrated balance between pro- and anticoagulant forces in plasma, distinguishes hypo- and hypercoagulable states in clinical populations [76], and has been associated with both initial and RVTE risk [77, 78]. These findings support the use of ETP as a clinically meaningful surrogate marker of coagulation perturbation in both mechanistic modeling and risk stratification contexts.

However, reliance on this scalar endpoint inevitably discards temporal information embedded in the thrombin generation curve. Dynamic features such as peak thrombin, lag time, time-to-peak, post-peak decay behavior, and maximum thrombin generation rate represent distinct physiological phases of initiation, amplification, and inhibition within the coagulation cascade. Incorporating these time-dependent characteristics in future surrogate frameworks may provide complementary mechanistic insight and potentially enhance discrimination and biological interpretability beyond what can be achieved with ETP alone.

The focus on ETP in the present study was motivated by its strong clinical relevance and interpretability. Nevertheless, future investigations should directly compare scalar ETP-based surrogates with multi-output or time-series architectures. Potential extensions include (i) joint prediction of multiple curve-derived metrics (e.g., peak thrombin and lag time), (ii) vector-valued surrogate models that simultaneously approximate several kinetic summaries, or (iii) sequence-based neural architectures capable of reconstructing the full thrombin generation curve. Moreover, incorporating experimental thrombin generation assay data would strengthen the biological grounding of the framework by enabling training and validation against empirically observed full-curve dynamics rather than relying solely on theoretical ODE-derived outputs.

Accordingly, the present findings demonstrate the feasibility and robustness of ANN surrogate modeling for scalar ETP prediction, while highlighting multi-output and full-curve modeling as important next steps toward enhanced mechanistic resolution and clinical translation.

Finally, the substantial computational acceleration achieved here enables a far greater number of exploratory and validation experiments within feasible time constraints. This makes iterative feature selection and kinetic parameter refinement strategies practically viable, increasing the potential to validate and refine the relationships among clinical features, kinetic parameters, ETP, and clinical outcomes. This enhanced experimental flexibility strengthens both methodological rigor and translational applicability of the proposed hybrid framework.

5 Conclusion

This study demonstrates that computationally intensive ODE-based coagulation models can be effectively replaced by ANN surrogates within hybrid optimization frameworks for RVTE prediction, yielding substantial efficiency gains without compromising predictive fidelity. Among the 14 ML algorithms evaluated, the ANN surrogate provided the best balance between accuracy and computational speed, reducing runtime by more than 99% while maintaining a relative accuracy of 97.97% under GWO optimization. This acceleration removes a major practical barrier associated with repeated ODE evaluations in high-dimensional optimization settings.

Comparative analyses across ML models and MOAs revealed that swarm-based optimizers, particularly GWO and PSO, consistently achieved superior performance in terms of fitness, discrimination, calibration, and robustness. Their effectiveness persisted across multiple stochastic runs and statistical comparisons, highlighting their suitability for navigating the highly nonconvex and multimodal objective landscape inherent to the hybrid framework. While ANN-GWO was statistically superior to most alternative MOA combinations and comparable only to ANN-PSO, no significant performance differences were observed between ANN-GWO and boosting-based models optimized with GWO. Nonetheless, the ANN surrogate achieved markedly lower RMSE values (below 0.3 in both training and testing sets) and lower residuals, confirming its strong approximation capacity relative to other ML approaches.

The optimal ANN architecture (41-7-10-1 neurons with tansig–logsig–purelin activation functions) provided a compact yet expressive representation of the underlying kinetic dynamics. Importantly, the synergy between ANN surrogate and swarm-based optimizers enabled reliable patient-specific risk stratification while preserving mechanistic interpretability through kinetic parameter analysis. Beyond predictive performance, the surrogate framework has the potential to facilitate deeper exploration of clinical–mechanistic relationships, sensitivity patterns, and robustness under stochastic variability, analyses that would be computationally prohibitive with the original ODE solver.

Overall, the findings support a practical and scalable strategy for the proposed hybrid mechanistic–data-driven modeling: replace stiff ODE systems with high-fidelity ANN surrogates and perform optimization using robust swarm-based algorithms such as GWO or PSO. Although developed in the context of RVTE prediction, the proposed surrogate-assisted optimization framework has broader applicability to other computationally intensive models. The approach can be extended to related hemostatic conditions, including broader VTE phenotypes and inherited bleeding disorders such as hemophilia, where dynamic coagulation modeling plays a central role.

Importantly, the substantial computational acceleration achieved in this study enables more intensive and systematic model development strategies. Iterative feedforward and feedback procedures for clinical feature selection and kinetic parameter refinement become computationally feasible, allowing deeper exploration of the relationships between clinical variables and kinetic parameters, as well as their joint influence on ETP and clinical outcomes. This expanded modeling capacity enhances both mechanistic insight and translational potential, supporting more refined and clinically informative hybrid modeling frameworks.

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Data availability The data are not publicly available because they contain information that could compromise the privacy of research participants.

Declarations

Conflict of interest The authors have no competing interests to declare that are relevant to the content of this article.

Ethical approval Ethical approval of the patient data used in this work was obtained and approved by the Comitê de Ética da Universidade Federal de Campinas (Approval number: 88970218.0.0000.5404).

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Authors and Affiliations

Mohamad Al Bannoud[1,4] · Tiago Dias Martins[2] · Silmara Aparecida de Lima Montalvão[3,4] · Joyce Maria Annichino-Bizzacchi[3,4] · Rubens Maciel Filho[1,4] · Maria Regina Wolf Maciel[1]

✉ Mohamad Al Bannoud
mohamad.bannoud@unifesp.br

1 School of Chemical Engineering, Laboratory of Optimization, Design, and Advanced Control, Universidade Estadual de Campinas, Campinas, Campinas, São Paulo, Brazil

2 Departamento de Engenharia Química, Instituto de Ciências Ambientais, Químicas e Farmacêuticas, Universidade Federal de São Paulo, Diadema, São Paulo, Brazil

3 Hematology and Hemotherapy Center, University of Campinas/ Hemocentro-Unicamp, Instituto Nacional de Ciência e Tecnologia do Sangue, Campinas, São Paulo, Brazil

4 Centro de Doenças Tromboembólicas (CCT), Centro de Hematologia e Hemoterapia (HEMOCENTRO), Universidade Estadual de Campinas, Campinas, São Paulo, Brazil